

Stellar Atmospheres

- 9.1 *The Description of the Radiation Field*
- 9.2 *Stellar Opacity*
- 9.3 *Radiative Transfer*
- 9.4 *The Transfer Equation*
- 9.5 *The Profiles of Spectral Lines*

9.1 ■ THE DESCRIPTION OF THE RADIATION FIELD

The light that astronomers receive from a star comes from the star's atmosphere, the layers of gas overlying the opaque interior. A flood of photons pours from these layers, releasing the energy produced by the thermonuclear reactions, gravitational contraction, and cooling in the star's center. The temperature, density, and composition of the atmospheric layers from which these photons escape determine the features of the star's spectrum. To interpret the observed spectral lines properly, we must describe how light travels through the gas that makes up a star.

The Specific and Mean Intensities

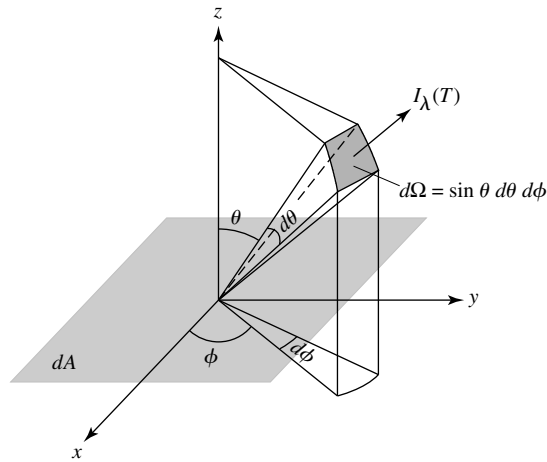
Figure 9.1 shows a ray of light with a wavelength between λ and $\lambda + d\lambda$ passing through a surface of area dA at an angle θ into a cone of solid angle $d\Omega$.¹ The angle θ is measured from the direction perpendicular to the surface, so $dA \cos \theta$ is the area dA projected onto a plane perpendicular to the direction in which the radiation is traveling. Defining

$$E_\lambda \equiv \frac{\partial E}{\partial \lambda},$$

$E_\lambda d\lambda$ is assumed to be the amount of energy that these rays carry into the cone in a time interval dt . Then the **specific intensity** of the rays is defined as

$$I_\lambda \equiv \frac{\partial I}{\partial \lambda} \equiv \frac{E_\lambda d\lambda}{d\lambda dt dA \cos \theta d\Omega}. \quad (9.1)$$

¹The surface is a mathematical location in space and is not necessarily a real physical surface. The concept of a solid angle and its units of steradians (sr) was discussed in Section 6.1.

FIGURE 9.1 Intensity I_λ .

Although the energy $E_\lambda d\lambda$ in the numerator is vanishingly small, the differentials in the denominator are also vanishingly small, so the ratio approaches a limiting value of I_λ . The specific intensity is usually referred to simply as the *intensity*. Thus, in spherical coordinates,

$$E_\lambda d\lambda = I_\lambda d\lambda dt dA \cos \theta d\Omega = I_\lambda d\lambda dt dA \cos \theta \sin \theta d\theta d\phi \quad (9.2)$$

is the amount of electromagnetic radiation energy having a wavelength between λ and $\lambda + d\lambda$ that passes in time dt through the area dA into a solid angle $d\Omega = \sin \theta d\theta d\phi$. The specific intensity therefore has units of $\text{W m}^{-3} \text{sr}^{-1}$.² The Planck function B_λ , Eq. (3.22), is an example of the specific intensity for the special case of blackbody radiation. In general, however, the energy of the light need not vary with wavelength in the same way as it does for blackbody radiation. Later we will see under what circumstances we may set $I_\lambda = B_\lambda$.

Imagine a light ray of intensity I_λ as it propagates through a vacuum. Because I_λ is defined in the limit $d\Omega \rightarrow 0$, the energy of the ray *does not spread out (or diverge)*. The intensity is therefore *constant* along any ray traveling through empty space.

In general, the specific intensity I_λ does vary with direction, however. The **mean intensity** of the radiation is found by integrating the specific intensity over all directions and dividing the result by 4π sr, the solid angle enclosed by a sphere, to obtain an average value of I_λ . In spherical coordinates, this average value is³

$$\langle I_\lambda \rangle \equiv \frac{1}{4\pi} \int I_\lambda d\Omega = \frac{1}{4\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_\lambda \sin \theta d\theta d\phi. \quad (9.3)$$

For an isotropic radiation field (one with the same intensity in all directions), $\langle I_\lambda \rangle = I_\lambda$. Blackbody radiation is isotropic and has $\langle I_\lambda \rangle = B_\lambda$.

²Recall from Section 3.5 that W m^{-3} indicates an energy per second per unit area *per unit wavelength interval*, $\text{W m}^{-2} \text{m}^{-1}$, *not* an energy per second per unit volume.

³Many texts refer to the average intensity as J_λ instead of $\langle I_\lambda \rangle$. However, in this text the notation $\langle I_\lambda \rangle$ has been selected to explicitly illustrate the average nature of the quantity.

The Specific Energy Density

To determine how much energy is contained within the radiation field, we can use a “trap” consisting of a small cylinder of length dL , open at both ends, with perfectly reflecting walls inside; see Fig. 9.2. Light entering the trap at one end travels and (possibly) bounces back and forth until it exits the other end of the trap. The energy inside the trap is the same as what would be present at that location if the trap were removed. The radiation that enters the trap at an angle θ travels through the trap in a time $dt = dL/(c \cos \theta)$. Thus the amount of energy inside the trap with a wavelength between λ and $\lambda + d\lambda$ that is due to the radiation that enters at angle θ is

$$E_\lambda d\lambda = I_\lambda d\lambda dt dA \cos \theta d\Omega = I_\lambda d\lambda dA d\Omega \frac{dL}{c}.$$

The quantity $dA dL$ is just the volume of the trap, so the **specific energy density** (energy per unit volume having a wavelength between λ and $\lambda + d\lambda$) is found by dividing $E_\lambda d\lambda$ by $dL dA$, integrating over all solid angles, and using Eq. (9.3):

$$\begin{aligned} u_\lambda d\lambda &= \frac{1}{c} \int I_\lambda d\lambda d\Omega \\ &= \frac{1}{c} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_\lambda d\lambda \sin \theta d\theta d\phi \\ &= \frac{4\pi}{c} \langle I_\lambda \rangle d\lambda. \end{aligned} \tag{9.4}$$

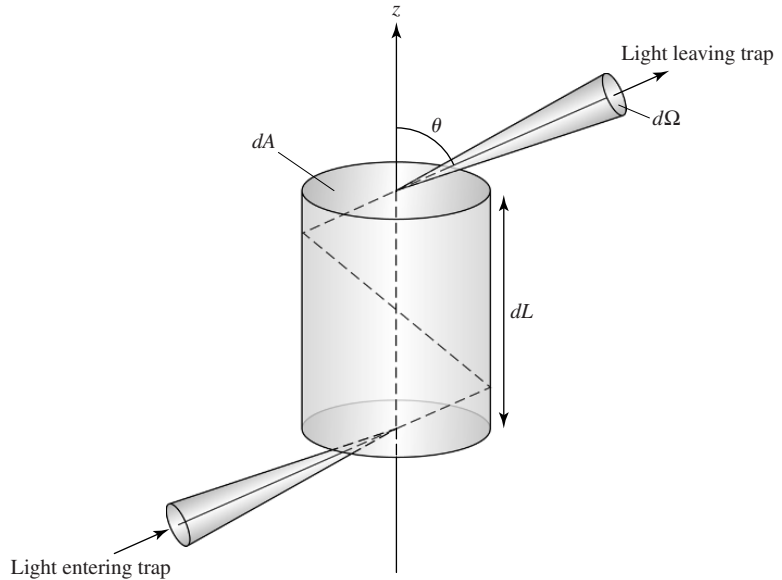


FIGURE 9.2 Cylindrical “trap” for measuring energy density u_λ .

For an isotropic radiation field, $u_\lambda d\lambda = (4\pi/c)I_\lambda d\lambda$, and for blackbody radiation,

$$u_\lambda d\lambda = \frac{4\pi}{c} B_\lambda d\lambda = \frac{8\pi hc/\lambda^5}{e^{hc/\lambda kT} - 1} d\lambda. \quad (9.5)$$

At times it may be more useful to express the blackbody energy density in terms of the frequency, ν , of the light by employing Eq. (3.24):

$$u_\nu d\nu = \frac{4\pi}{c} B_\nu d\nu = \frac{8\pi h\nu^3/c^3}{e^{h\nu/kT} - 1} d\nu. \quad (9.6)$$

Thus $u_\nu d\nu$ is the energy per unit volume with a frequency between ν and $\nu + d\nu$.

The total energy density, u , is found by integrating over all wavelengths or over all frequencies:

$$u = \int_0^\infty u_\lambda d\lambda = \int_0^\infty u_\nu d\nu.$$

For blackbody radiation ($I_\lambda = B_\lambda$), Eq. (3.28) shows that

$$u = \frac{4\pi}{c} \int_0^\infty B_\lambda(T) d\lambda = \frac{4\sigma T^4}{c} = aT^4, \quad (9.7)$$

where $a \equiv 4\sigma/c$ is known as the *radiation constant* and has the value

$$a = 7.565767 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}.$$

The Specific Radiative Flux

Another quantity of interest is F_λ , the **specific radiative flux**. $F_\lambda d\lambda$ is the *net* energy having a wavelength between λ and $\lambda + d\lambda$ that passes each second through a unit area in the direction of the z -axis:

$$F_\lambda d\lambda = \int I_\lambda d\lambda \cos \theta d\Omega = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_\lambda d\lambda \cos \theta \sin \theta d\theta d\phi. \quad (9.8)$$

The factor of $\cos \theta$ determines the z -component of a light ray and allows the cancelation of oppositely directed rays. For an isotropic radiation field there is no net transport of energy, and so $F_\lambda = 0$.

Both the radiative flux and the specific intensity describe the light received from a celestial source, and you may wonder which of these quantities is actually measured by a telescope's photometer, pointed at the source of light. The answer depends on whether the source is resolved by the telescope. Figure 9.3(a) shows a source of light, uniform over its entire surface,⁴ that is resolved by the telescope; the angle θ subtended by the source as a whole is much larger than $\theta_{\min}$, the smallest angle resolvable according to Rayleigh's

⁴The assumption of a uniform light source precludes dimming effects such as limb darkening, which will be discussed later.

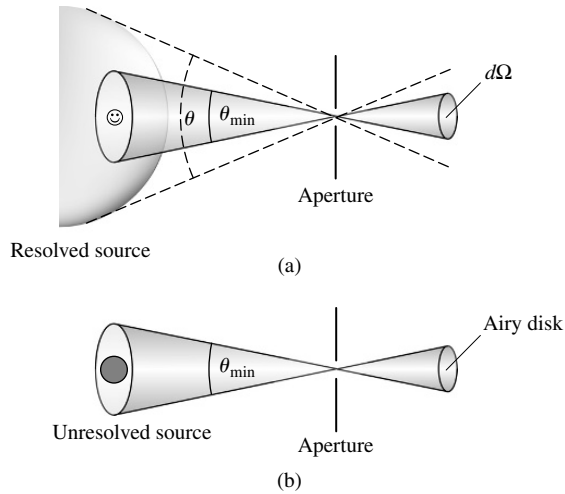


FIGURE 9.3 The measurement of (a) the specific intensity for a resolved source and (b) the radiative flux for an unresolved source. Note that any object with an angular resolution smaller than $\theta_{\min}$ on the surface of the resolved source (such as a surface feature on a planet) remains unresolved.

criterion. In this case, what is being measured is the *specific intensity*, the amount of energy per second passing through the aperture area into the solid angle $\Omega_{\min}$ defined by $\theta_{\min}$. For example, at a wavelength of 501 nm, the measured value of the specific intensity at the center of the Sun's disk is

$$I_{501} = 4.03 \times 10^{13} \text{ W m}^{-3} \text{ sr}^{-1}.$$

Now imagine that the source is moved twice as far away. According to the inverse square law for light, Eq. (3.2), there will be only $(1/2)^2 = 1/4$ as much energy received from each square meter of the source. If the source is still resolved, however, then the amount of source area that contributes energy to the solid angle $\Omega_{\min}$ has increased by a factor of 4, resulting in the same amount of energy reaching each square meter of the detector. The specific intensity of light rays from the source is thus measured to be constant.⁵

However, it is the *radiative flux* that is measured for an unresolved source. As the source recedes farther and farther, it will eventually subtend an angle θ smaller than $\theta_{\min}$, and it can no longer be resolved by the telescope. When $\theta < \theta_{\min}$, the energy received from the entire source will disperse throughout the diffraction pattern (the Airy disk and rings; recall Section 6.1) determined by the telescope's aperture. Because the light arriving at the detector leaves the surface of the source at all angles [see Fig. 9.3(b)], the detector is effectively integrating the specific intensity over all directions. This is just the definition of the radiative flux, Eq. (9.8). As the distance r to the source increases further, the amount of energy falling within the Airy disk (and consequently the value of the radiative flux) decreases as $1/r^2$, as expected.

⁵We encountered this argument in the statement in Section 6.1 that the image and object intensities of a resolved object are the same.

Radiation Pressure

Because a photon possesses an energy E , Einstein's relativistic energy equation (Eq. 4.48) tells us that even though it is massless, a photon also carries a momentum of $p = E/c$ and thus can exert a **radiation pressure**. This radiation pressure can be derived in the same way that gas pressure is found for molecules bouncing off a wall. Figure 9.4 shows photons reflected at an angle θ from a perfectly reflecting surface of area dA into a solid angle $d\Omega$. Because the angle of incidence equals the angle of reflection, the solid angles shown for the incident and reflected photons are the same size and inclined by the same angle θ on opposing sides of the z -axis. The change in the z -component of the momentum of photons with wavelengths between λ and $\lambda + d\lambda$ that are reflected from the area dA in a time interval dt is

$$\begin{aligned}
 dp_\lambda d\lambda &= [(p_\lambda)_{\text{final},z} - (p_\lambda)_{\text{initial},z}] d\lambda \\
 &= \left[\frac{E_\lambda \cos \theta}{c} - \left(-\frac{E_\lambda \cos \theta}{c} \right) \right] d\lambda \\
 &= \frac{2 E_\lambda \cos \theta}{c} d\lambda \\
 &= \frac{2}{c} I_\lambda d\lambda dt dA \cos^2 \theta d\Omega,
 \end{aligned}$$

where the last expression was obtained from Eq. (9.2). Dividing dp_λ by dt and dA gives $(dp_\lambda/dt)/dA$. But from Newton's second and third laws, $-dp_\lambda/dt$ is the force exerted by the photons on the area dA , although we will ignore the minus sign, which merely says that the force is in the $-z$ -direction. Thus the radiation pressure is the force per unit area, $(dp_\lambda/dt)/dA$, produced by the photons within the solid angle $d\Omega$. Integrating over the hemisphere of all incident directions results in $P_{\text{rad},\lambda} d\lambda$, the radiation pressure exerted by

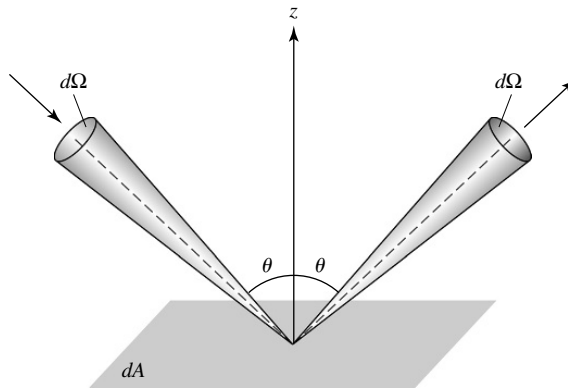


FIGURE 9.4 Radiation pressure produced by incident photons from the solid angle $d\Omega$.

those photons having a wavelength between λ and $\lambda + d\lambda$:

$$\begin{aligned} P_{\text{rad},\lambda} d\lambda &= \frac{2}{c} \int_{\text{hemisphere}} I_{\lambda} d\lambda \cos^2 \theta d\Omega \quad (\text{reflection}) \\ &= \frac{2}{c} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda} d\lambda \cos^2 \theta \sin \theta d\theta d\phi. \end{aligned}$$

Just as the pressure of a gas exists throughout the volume of the gas and not just at the container walls, the radiation pressure of a “photon gas” exists everywhere in the radiation field. Imagine removing the reflecting surface dA in Fig. 9.4 and replacing it with a mathematical surface. The incident photons will now keep on going through dA ; instead of reflected photons, photons will be streaming through dA from the other side. Thus, for an *isotropic radiation field*, there will be no change in the expression for the radiation pressure if the leading factor of 2 (which originated in the change in momentum upon reflection of the photons) is removed and the angular integration is extended over all solid angles:

$$P_{\text{rad},\lambda} d\lambda = \frac{1}{c} \int_{\text{sphere}} I_{\lambda} d\lambda \cos^2 \theta d\Omega \quad (\text{transmission}) \quad (9.9)$$

$$\begin{aligned} &= \frac{1}{c} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_{\lambda} d\lambda \cos^2 \theta \sin \theta d\theta d\phi \\ &= \frac{4\pi}{3c} I_{\lambda} d\lambda \quad (\text{isotropic radiation field}). \end{aligned} \quad (9.10)$$

However, it may be that the radiation field is *not* isotropic. In that case, Eq. (9.9) for the radiation pressure is still valid but the pressure depends on the orientation of the mathematical surface dA .

The total radiation pressure produced by photons of all wavelengths is found by integrating Eq. (9.10):

$$P_{\text{rad}} = \int_0^{\infty} P_{\text{rad},\lambda} d\lambda.$$

For blackbody radiation, it is left as a problem to show that

$$P_{\text{rad}} = \frac{4\pi}{3c} \int_0^{\infty} B_{\lambda}(T) d\lambda = \frac{4\sigma T^4}{3c} = \frac{1}{3} a T^4 = \frac{1}{3} u. \quad (9.11)$$

Thus the **blackbody radiation pressure** is one-third of the energy density. (For comparison, the pressure of an ideal monatomic gas is two-thirds of its energy density.)

9.2 ■ STELLAR OPACITY

The classification of stellar spectra is an ongoing process. Even the most basic task, such as finding the “surface”⁶ temperature of a particular star, is complicated by the fact that stars are not actually blackbodies. The Stefan–Boltzmann relation, in the form of Eq. (3.17), defines a star’s effective temperature, but some effort is required to obtain a more accurate value of the “surface” temperature.⁷ Figure 9.5 shows that the Sun’s spectrum deviates substantially from the shape of the blackbody Planck function, B_λ , because solar absorption lines remove light from the Sun’s continuous spectrum at certain wavelengths. The decrease in intensity produced by the dense series of metallic absorption lines in the solar spectrum is especially effective; this effect is called **line blanketing**. In other wavelength regimes (e.g., X-ray and UV), emission lines may augment the intensity of the continuous spectrum.

Temperature and Local Thermodynamic Equilibrium

Although we often think in terms of *the temperature* at a particular location, there are actually many different measures of temperature within a star, defined according to the physical process being described:

- The **effective temperature**, which is obtained from the Stefan–Boltzmann law (Eq. 3.17), is uniquely defined for a specific level within a star and is an important global descriptor of that star.
- The **excitation temperature** is defined by the Boltzmann equation (8.6).
- The **ionization temperature** is defined by the Saha equation (8.8).

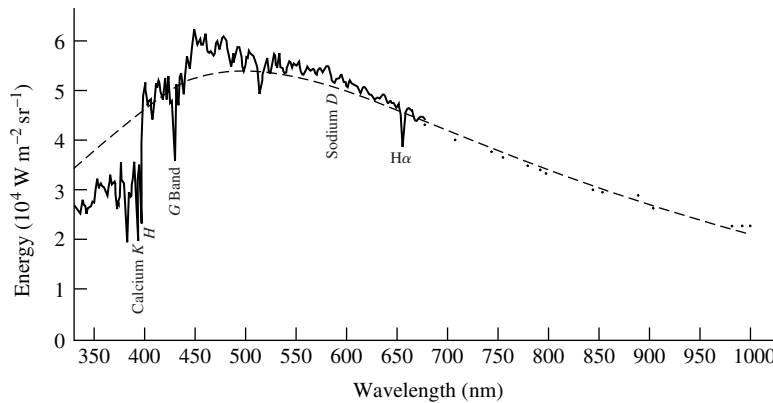


FIGURE 9.5 The spectrum of the Sun in 2 nm wavelength intervals. The dashed line is the curve of an ideal blackbody having the Sun’s effective temperature. (Figure adapted from Aller, *Atoms, Stars, and Nebulae*, Third Edition, Cambridge University Press, New York, 1991.)

⁶The “surface” of a star is defined as the region where the emergent visual continuum forms, namely the *photosphere* (see Section 11.2).

⁷See Böhm-Vitense (1981) for more details concerning the determination of temperatures.

- The **kinetic temperature** is contained in the Maxwell–Boltzmann distribution, Eq. (8.1).
- The **color temperature** is obtained by fitting the shape of a star’s continuous spectrum to the Planck function, Eq. (3.22).

With the exception of the effective temperature, the remaining temperatures apply to any location within the star and vary according to the conditions of the gas. Although defined differently, the excitation temperature, the ionization temperature, the kinetic temperature, and the color temperature are the same for the simple case of a gas confined within an “ideal box.” The confined gas particles and blackbody radiation will come into equilibrium, individually and with each other, and can be described by a single well-defined temperature. In such a steady-state condition, no net flow of energy through the box or between the matter and the radiation occurs. Every process (e.g., the absorption of a photon) occurs at the same rate as its inverse process (e.g., the emission of a photon). This condition is called **thermodynamic equilibrium**.

However, a star cannot be in perfect thermodynamic equilibrium. A net outward flow of energy occurs through the star, and the temperature, however it is defined, varies with location. Gas particles and photons at one position in the star may have arrived there from other regions, either hotter or cooler (in other words, there is no “ideal box”). The distribution in particle speeds and photon energies thus reflects a range of temperatures. As the gas particles collide with one another and interact with the radiation field by absorbing and emitting photons, the description of the processes of excitation and ionization becomes quite complex. However, the idealized case of a single temperature can still be employed if the distance over which the temperature changes significantly is large compared with the distances traveled by the particles and photons between collisions (their *mean free paths*). In this case, referred to as **local thermodynamic equilibrium** (LTE), the particles and photons cannot escape the local environment and so are effectively confined to a limited volume (an approximated “box”) of nearly constant temperature.

Example 9.2.1. The photosphere is the surface layer of the Sun’s atmosphere where the photons can escape into space (see page 253 and Section 11.2). According to a model solar atmosphere (see Cox, page 348), the temperature in one region of the photosphere varies from 5580 K to 5790 K over a distance of 25.0 km. The characteristic distance over which the temperature varies, called the *temperature scale height*, H_T , is given by

$$H_T \equiv \frac{T}{|dT/dr|} = \frac{5685 \text{ K}}{(5790 \text{ K} - 5580 \text{ K})/(25.0 \text{ km})} = 677 \text{ km},$$

where the average temperature has been used for the value of T .

How does the temperature scale height of 677 km compare with the average distance traveled by an atom before hitting another atom? The density of the photosphere at that level is about $\rho = 2.1 \times 10^{-4} \text{ kg m}^{-3}$, consisting primarily of neutral hydrogen atoms in the ground state. Assuming a pure hydrogen gas for convenience, the number of hydrogen

continued

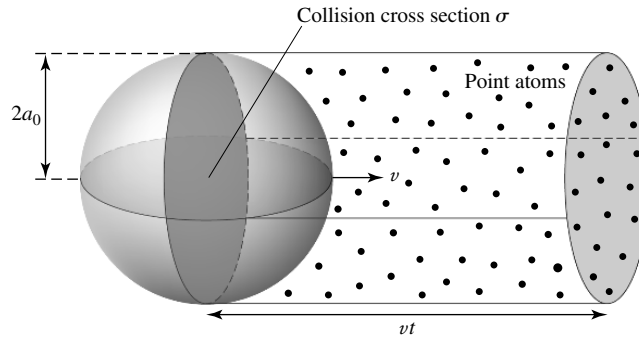


FIGURE 9.6 Mean free path, ℓ , of a hydrogen atom.

atoms per cubic meter is roughly

$$n = \frac{\rho}{m_H} = 1.25 \times 10^{23} \text{ m}^{-3},$$

where m_H is the mass of a hydrogen atom. In an approximate sense, two of these atoms will “collide” if their centers pass within two Bohr radii, $2a_0$, of each other.⁸ As shown in Fig. 9.6, we may consider the equivalent problem of a single atom of radius $2a_0$ moving with speed v through a collection of stationary points that represent the centers of the other atoms. In an amount of time t , this atom has moved a distance vt and has swept out a cylindrical volume $V = \pi(2a_0)^2 vt = \sigma vt$, where $\sigma \equiv \pi(2a_0)^2$ is the collision **cross section** of the atom in this classical approximation.⁹ Within this volume V are $nV = n\sigma vt$ point atoms with which the moving atom has collided. Thus the average distance traveled between collisions is

$$\ell = \frac{vt}{n\sigma vt} = \frac{1}{n\sigma}. \quad (9.12)$$

The distance ℓ is the **mean free path** between collisions.¹⁰ For a hydrogen atom,

$$\sigma = \pi(2a_0)^2 = 3.52 \times 10^{-20} \text{ m}^2.$$

Thus the mean free path in this situation is

$$\ell = \frac{1}{n\sigma} = 2.27 \times 10^{-4} \text{ m}.$$

The mean free path is several billion times smaller than the temperature scale height. As a result, the atoms in the gas see an essentially constant kinetic temperature between collisions. They are effectively confined within a limited volume of space in the photosphere. Of course this cannot be true for the photons as well, since the Sun’s photosphere is the visible layer

⁸This treats the atoms as solid spheres, a classical approximation to the quantum atom.

⁹The concept of *cross section*, which will be discussed in more detail in Section 10.3, actually represents a probability of particle interactions but has units of cross-sectional area.

¹⁰A more careful calculation, using a Maxwellian velocity distribution for all of the atoms, results in a mean free path that is smaller by a factor of $\sqrt{2}$.

of the solar surface that we observe from Earth. Thus, by the very definition of photosphere, the photons must be able to escape freely into space. To say more about the photon mean free path and the concept of LTE, and to better understand the solar spectrum shown in Fig. 9.5, we must examine the interaction of particles and photons in some detail.

The Definition of Opacity

We now turn to a consideration of a beam of parallel light rays traveling through a gas. Any process that removes photons from a beam of light will be collectively termed **absorption**. In this sense then, absorption includes the *scattering* of photons (such as Compton scattering, discussed in Section 5.2) as well as the true absorption of photons by atomic electrons making upward transitions. In sufficiently cool gases, molecular energy-level transitions may also occur and must be included.

The change in the intensity, dI_λ , of a ray of wavelength λ as it travels through a gas is proportional to its intensity, I_λ , the distance traveled, ds , and the density of the gas, ρ . That is,

$$dI_\lambda = -\kappa_\lambda \rho I_\lambda ds. \quad (9.13)$$

The distance s is measured along the path traveled by the beam and increases in the direction that the beam travels; the minus sign in Eq. (9.13) shows that the intensity *decreases* with distance due to the absorption of photons. The quantity κ_λ is called the **absorption coefficient**, or **opacity**, with the λ subscript implicitly indicating that the opacity is wavelength-dependent (κ_λ is sometimes referred to as a *monochromatic opacity*). The opacity is the cross section for absorbing photons of wavelength λ per unit mass of stellar material and has units of $\text{m}^2 \text{kg}^{-1}$. In general, the opacity of a gas is a function of its composition, density, and temperature.¹¹

Example 9.2.2. Consider a beam of light traveling through a gas with initial intensity $I_{\lambda,0}$ at $s = 0$. The final intensity $I_{\lambda,f}$ after the light has traveled a distance s may be found by integrating Eq. (9.13):

$$\int_{I_{\lambda,0}}^{I_{\lambda,f}} \frac{dI_\lambda}{I_\lambda} = - \int_0^s \kappa_\lambda \rho ds.$$

This leads to

$$I_\lambda = I_{\lambda,0} e^{-\int_0^s \kappa_\lambda \rho ds}, \quad (9.14)$$

where the f subscript has been dropped. For the specific case of a uniform gas of constant opacity and density,

$$I_\lambda = I_{\lambda,0} e^{-\kappa_\lambda \rho s}.$$

continued

¹¹Note that there is some inconsistency in the terminology; some authors refer to *opacity* as the inverse of the mean free path of the photons.

For pure absorption (with emission processes neglected), there is no way of replenishing the photons lost from the beam. The intensity declines exponentially, falling by a factor of e^{-1} over a characteristic distance of $\ell = 1/\kappa_\lambda \rho$. In the solar photosphere where the density is approximately $\rho = 2.1 \times 10^{-4} \text{ kg m}^{-3}$, the opacity (at a wavelength of 500 nm) is $\kappa_{500} = 0.03 \text{ m}^2 \text{ kg}^{-1}$. Thus the characteristic distance traveled by a photon before being removed from the beam at this level in the photosphere is

$$\ell = \frac{1}{\kappa_{500} \rho} = 160 \text{ km.}$$

Recalling Example 9.2.1, this distance is comparable to the temperature scale height $H_T = 677 \text{ km}$. This implies that the photospheric photons do not see a constant temperature, and so local thermodynamic equilibrium (LTE) is not strictly valid in the photosphere. The temperature of the regions from which the photons have traveled will be somewhat different from the local kinetic temperature of the gas. Although LTE is a commonly invoked assumption in stellar atmospheres, it must be used with caution.

Optical Depth

For scattered photons, the characteristic distance ℓ is in fact the mean free path of the photons. From Eq. (9.12),

$$\ell = \frac{1}{\kappa_\lambda \rho} = \frac{1}{n \sigma_\lambda}.$$

Both $\kappa_\lambda \rho$ and $n \sigma_\lambda$ can be thought of as the fraction of photons scattered per meter of distance travelled. Note that the mean free path is different for photons of different wavelengths.

It is convenient to define an **optical depth**, τ_λ , back along a light ray by

$$\boxed{d\tau_\lambda = -\kappa_\lambda \rho ds,} \quad (9.15)$$

where s is the distance measured along the photon's path in its direction of motion (when observing the light from a star, we are *looking back* along the path traveled by the photon; see Fig. 9.7). The difference in optical depth between a light ray's initial position ($s = 0$) and its final position after traveling a distance s is

$$\Delta\tau_\lambda = \tau_{\lambda,f} - \tau_{\lambda,0} = -\int_0^s \kappa_\lambda \rho ds. \quad (9.16)$$

Note that $\Delta\tau_\lambda < 0$; as the light approaches an observer, it is traveling through material with diminishing optical depth. The outermost layers of a star may be taken to be at $\tau_\lambda = 0$ for all wavelengths, after which the light travels unimpeded to observers on Earth. With this definition of $\tau_\lambda = 0$, Eq. (9.16) gives the initial optical depth, $\tau_{\lambda,0}$, of a ray of light that

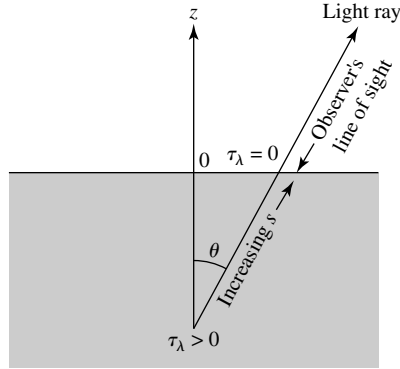


FIGURE 9.7 Optical depth τ_λ measured back along a ray's path.

traveled a distance s to reach the top of the atmosphere:

$$\begin{aligned}
 0 - \tau_{\lambda,0} &= - \int_0^s \kappa_\lambda \rho \, ds \\
 \tau_\lambda &= \int_0^s \kappa_\lambda \rho \, ds.
 \end{aligned} \tag{9.17}$$

The “0” subscript has been dropped with the understanding that τ_λ is the optical depth of the ray's initial position, a distance s ($s > 0$) from the top of the atmosphere.

Combining Eq. (9.17) with Eq. (9.14) of Example 9.2.2 for the case of pure absorption, we find that the decline in the intensity of a ray that travels through a gas from an optical depth τ_λ to reach the observer is given by

$$I_\lambda = I_{\lambda,0} e^{-\tau_\lambda}. \tag{9.18}$$

Thus, if the optical depth of the ray's starting point is $\tau_\lambda = 1$, the intensity of the ray will decline by a factor of e^{-1} before escaping from the star. *The optical depth may be thought of as the number of mean free paths from the original position to the surface, as measured along the ray's path.* As a result, we typically see no deeper into an atmosphere at a given wavelength than $\tau_\lambda \approx 1$. Of course, for pure absorption the intensity of the ray declines exponentially regardless of its direction of travel through the gas. But we can observe only those rays traveling toward us, and this is reflected in our choice of $\tau_\lambda = 0$ at the top of the atmosphere. Other choices of where $\tau_\lambda = 0$ may be more useful in some situations.

If $\tau_\lambda \gg 1$ for a light ray passing through a volume of gas, the gas is said to be *optically thick*; if $\tau_\lambda \ll 1$, the gas is *optically thin*. Because the optical depth varies with wavelength, a gas may be optically thick at one wavelength and optically thin at another. For example, Earth's atmosphere is optically thin at visible wavelengths (we can see the stars), but optically thick at X-ray wavelengths; e.g., recall Fig. 6.25.

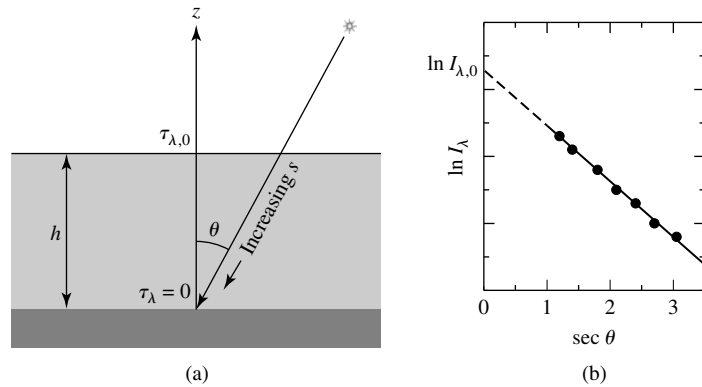


FIGURE 9.8 (a) A light ray entering Earth's atmosphere at an angle θ . (b) $\ln I_\lambda$ vs. $\sec \theta$.

Example 9.2.3. In Section 5.2, we stated that measurements of a star's radiative flux and apparent magnitude are routinely corrected for the absorption of light by Earth's atmosphere. Figure 9.8(a) shows a ray of intensity $I_{\lambda,0}$ entering Earth's atmosphere at an angle θ and traveling to a telescope on the ground. The intensity of the light detected at the telescope is I_λ ; the problem is to determine the value of $I_{\lambda,0}$. If we take $\tau_\lambda = 0$ at the telescope and h to be the height of the atmosphere, then the optical depth of the light ray's path through the atmosphere may be found from Eq. (9.17). Using $ds = -dz/\cos \theta = -\sec \theta dz$ yields

$$\tau_\lambda = \int_0^s \kappa_\lambda \rho ds = - \int_h^0 \kappa_\lambda \rho \frac{dz}{\cos \theta} = \sec \theta \int_0^h \kappa_\lambda \rho dz = \tau_{\lambda,0} \sec \theta,$$

where $\tau_{\lambda,0}$ is the optical depth for a vertically traveling photon ($\theta = 0$). Substituting into Eq. (9.18), the intensity of the light received at the telescope is therefore given by

$$I_\lambda = I_{\lambda,0} e^{-\tau_{\lambda,0} \sec \theta}. \quad (9.19)$$

There are two unknowns in this equation, $I_{\lambda,0}$ and $\tau_{\lambda,0}$; neither can be determined by a single observation. However, as time passes and as Earth rotates on its axis, the angle θ will change, and a semilog graph of several measurements of the received intensity I_λ as a function of $\sec \theta$ can be made. As shown in Fig. 9.8(b), the *slope* of the best-fitting straight line is $-\tau_{\lambda,0}$. Extrapolating the best-fitting line to $\sec \theta = 0$ provides the value of $I_{\lambda,0}$ at the point where the line intercepts the I_λ -axis.¹² In this way, measurements of the specific intensity or radiative flux can be corrected for absorption by Earth's atmosphere.

General Sources of Opacity

The opacity of the stellar material is determined by the details of how photons interact with particles (atoms, ions, and free electrons). If the photon passes within σ_λ of the particle,

¹²Note that since $\sec \theta \geq 1$, the best-fitting straight line must be extrapolated to the mathematically unavailable value of 0.

where σ_λ is the particle's cross-sectional area (or effective target area), the photon may be either absorbed or scattered. In an absorption process, the photon ceases to exist and its energy is given up to the thermal energy of the gas. In a scattering process the photon continues on in a different direction. Both absorption and scattering can remove photons from a beam of light, and so contribute to the opacity, κ_λ , of the stellar material. If the opacity varies slowly with wavelength, it determines the star's continuous spectrum (or *continuum*). The dark absorption lines superimposed on the continuum are the result of a rapid variation in the opacity with wavelength (recall, for example, Figs. 8.2 and 8.3).

In general, there are four primary sources of opacity available for removing stellar photons from a beam. Each involves a change in the quantum state of an electron, and the terms *bound* and *free* are used to describe whether the electron is bound to an atom or ion in its initial and final states.

1. **Bound–bound transitions** (excitations and de-excitations) occur when an electron in an atom or ion makes a transition from one orbital to another. An electron can make an upward transition from a lower- to a higher-energy orbital when a photon of the appropriate energy is absorbed. Thus $\kappa_{\lambda,\text{bb}}$, the bound–bound opacity, is small except at those discrete wavelengths capable of producing an upward atomic transition. It is $\kappa_{\lambda,\text{bb}}$ that is responsible for forming the absorption lines in stellar spectra. The reverse process, emission, occurs when the electron makes a downward transition from a higher- to a lower-energy orbital.

If an electron absorbs a photon and then returns directly to its initial orbital (where it was before absorbing the photon), then a single photon is emitted in a random direction. The net result of this absorption–emission sequence is essentially a scattered photon. Otherwise, if the electron makes a transition to an orbital other than its initial one, the original photon is not recovered and the process is one of true absorption. If, while in its excited state, the atom or ion collides with a neighboring particle, collisional de-excitation may result. When this occurs, the energy lost by the atom or ion becomes a part of the thermal energy of the gas.

An important by-product of this absorption process is degrading of the average energy of the photons in the radiation field. For example, if one photon is absorbed but two photons are emitted as the electron cascades down to its initial orbital, then the average photon energy has been reduced by half. There is no simple equation for bound–bound transitions that describes all of the contributions to the opacity by individual spectral lines.

2. **Bound–free absorption**, also known as *photoionization*, occurs when an incident photon has enough energy to ionize an atom. The resulting free electron can have any energy, so any photon with a wavelength $\lambda \leq hc/\chi_n$, where χ_n is the ionization energy of the n th orbital, can remove an electron from an atom. Thus $\kappa_{\lambda,\text{bf}}$, the bound–free opacity, is one source of the continuum opacity. The cross section for the photoionization of a hydrogen atom in quantum state n by a photon of wavelength λ is

$$\sigma_{\text{bf}} = 1.31 \times 10^{-19} \frac{1}{n^5} \left(\frac{\lambda}{500 \text{ nm}} \right)^3 \text{ m}^2,$$

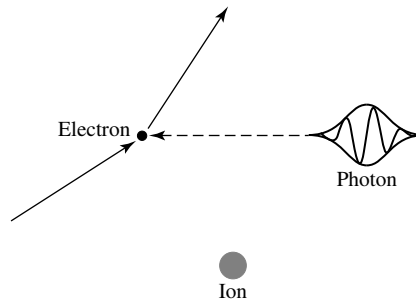


FIGURE 9.9 Free–free absorption of a photon.

which is comparable to the collision cross section for hydrogen found in Example 9.2.1. The inverse process of free–bound emission occurs when a free electron recombines with an ion, emitting one or more photons in random directions. As with bound–bound emission, this also contributes to reducing the average energy of the photons in the radiation field.

3. **Free–free absorption** is a scattering process, shown in Fig. 9.9, that takes place when a free electron in the vicinity of an ion absorbs a photon, causing the speed of the electron to increase. In this process the nearby ion is necessary in order to conserve both energy and momentum. (It is left as an exercise to show that an isolated free electron cannot absorb a photon.) Since this mechanism can occur for a continuous range of wavelengths, free–free opacity, $\kappa_{\lambda, \text{ff}}$, is another contributor to the continuum opacity. It may also happen that as it passes near an ion, the electron loses energy by emitting a photon, which causes the electron to slow down. This process of free–free emission is also known as *bremsstrahlung*, which means “braking radiation” in German.
4. **Electron scattering** is as advertised: A photon is scattered (*not absorbed*) by a free electron through the process of *Thomson scattering*. In this process, the electron can be thought of as being made to oscillate in the electromagnetic field of the photon. However, because the electron is tiny, it makes a poor target for an incident photon, resulting in a small cross section. The cross section for Thomson scattering has the same value for photons of all wavelengths:

$$\sigma_T = \frac{1}{6\pi\epsilon_0^2} \left(\frac{e^2}{m_e c^2} \right)^2 = 6.65 \times 10^{-29} \text{ m}^2. \quad (9.20)$$

This is typically two billion times smaller than the hydrogen cross section for photoionization, σ_{bf} . The small size of the Thomson cross section means that electron scattering is most effective as a source of opacity when the electron density is very high, which requires high temperature. In the atmospheres of the hottest stars (and in the interiors of all stars), where most of the gas is completely ionized, other sources of opacity that involve bound electrons are eliminated. In this high-temperature regime, the opacity due to electron scattering, κ_{es} , dominates the continuum opacity.

A photon may also be scattered by an electron that is loosely bound to an atomic nucleus. This result is called *Compton scattering* if the photon's wavelength is much smaller than the atom or *Rayleigh scattering* if the photon's wavelength is much larger. In Compton scattering, the change in the wavelength and energy of the scattered photon is very small (recall the discussion of the Compton wavelength on page 119), so Compton scattering is usually lumped together with Thomson scattering. The cross section for Rayleigh scattering from a loosely bound electron is smaller than the Thomson cross section; it is proportional to $1/\lambda^4$ and so decreases with increasing photon wavelength. Rayleigh scattering can be neglected in most atmospheres, but it is important in the UV for the extended envelopes of supergiant stars, and in cool main-sequence stars.¹³ The scattering of photons from small particles is also responsible for the reddening of starlight as it passes through interstellar dust; see Section 12.1.

Example 9.2.4. The energy of an electron in the $n = 2$ orbit of a hydrogen atom is given by Eq. (5.14):

$$E_2 = -\frac{13.6}{2^2} \text{ eV} = -3.40 \text{ eV}.$$

A photon must have an energy of at least $\chi_2 = 3.40 \text{ eV}$ to eject this electron from the atom. Thus any photon with a wavelength

$$\lambda \leq \frac{hc}{\chi_2} = 364.7 \text{ nm}$$

is capable of ionizing a hydrogen atom in the first excited state ($n = 2$). The opacity of the stellar material suddenly increases at wavelengths $\lambda \leq 364.7 \text{ nm}$, and the radiative flux measured for the star accordingly decreases. The abrupt drop in the continuous spectrum of a star at this wavelength, called the **Balmer jump**, is evident in the Sun's spectrum (Fig. 9.5). The size of the Balmer jump in hot stars depends on the fraction of hydrogen atoms that are in the first excited state. This fraction is determined by the temperature via the Boltzmann equation (Eq. 8.6), so a measurement of the size of the Balmer jump can be used to determine the temperature of the atmosphere. For cooler or very hot stars with other significant sources of opacity, the analysis is more complicated, but the size of the Balmer jump can still be used as a probe of atmospheric temperatures.

The wavelength 364.7 nm is right in the middle of the bandwidth of the ultraviolet (U) filter in the UBV system, described on page 75. As a result, the Balmer jump will tend to decrease the amount of light received in the bandwidth of the U filter and so *increase* both the ultraviolet magnitude U and the color index ($U - B$) observed for a star. This effect will be strongest when N_2/N_{total} , the fraction of all hydrogen atoms that are in the first

continued

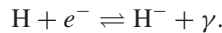
¹³Rayleigh scattering is also important in planetary atmospheres and is responsible for Earth's blue sky, for instance.

excited state, is a maximum. From Example 8.1.4, this occurs at a temperature of 9600 K, about the temperature of an A0 star on the main sequence. A careful examination of the color-color diagram in Fig. 3.11 reveals that this is indeed the spectral type at which the value of $U - B$ differs most from its blackbody value. The effect of line blanketing affects the measured color indices, making the star appear more red than a model blackbody star of the same effective temperature, and thus increasing the values of both $U - B$ and $B - V$.

Continuum Opacity and the H^- Ion

The primary source of the continuum opacity in the atmospheres of stars later than F0 is the photoionization of H^- ions. An H^- ion is a hydrogen atom that possesses an extra electron. Because of the partial shielding that the nucleus provides, a second electron can be loosely bound to the atom on the side of the ion opposite that of the first electron. In this position the second electron is closer to the positively charged nucleus than it is to the negatively charged electron. Therefore, according to Coulomb's law (Eq. 5.9), the net force on the extra electron is attractive.

The binding energy of the H^- ion is only 0.754 eV, compared with the 13.6 eV required to ionize the ground state hydrogen atom. As a result, any photon with energy in excess of the ionization energy can be absorbed by an H^- ion, liberating the extra electron; the remaining energy becomes kinetic energy. Conversely, an electron captured by a hydrogen atom to form H^- will release a photon corresponding to the kinetic energy lost by the electron together with the ion's binding energy,



Since 0.754 eV corresponds to a photon with a wavelength of 1640 nm, any photon with a wavelength less than that value can remove an electron from the ion (bound-free opacity). At longer wavelengths, H^- can also contribute to the opacity through free-free absorption. Consequently, H^- ions are an important source of continuum opacity for stars cooler than F0. However, the H^- ions become increasingly ionized at higher temperatures and therefore make less of a contribution to the continuum opacity. For stars of spectral types B and A, the photoionization of hydrogen atoms and free-free absorption are the main sources of the continuum opacity. At the even higher temperatures encountered for O stars, the ionization of atomic hydrogen means that electron scattering becomes more and more important, with the photoionization of helium also contributing to the opacity.

Molecules can survive in cooler stellar atmospheres and contribute to the bound-bound and bound-free opacities; the large number of discrete molecular absorption lines is an efficient impediment to the flow of photons. Molecules can also be broken apart into their constituent atoms by the absorption of photons in the process of *photodissociation*, which plays an important role in planetary atmospheres.

The total opacity is the sum of the opacities due to all of the preceding sources:

$$\kappa_\lambda = \kappa_{\lambda,bb} + \kappa_{\lambda,bf} + \kappa_{\lambda,ff} + \kappa_{es} + \kappa_{H^-}$$

(the H^- opacity is explicitly included because of its unique and critical contribution to the opacity in many stellar atmospheres, including our Sun). The total opacity depends not only

on the wavelength of the light being absorbed but also on the composition, density, and temperature of the stellar material.¹⁴

The Rosseland Mean Opacity

It is often useful to employ an opacity that has been averaged over all wavelengths (or frequencies) to produce a function that depends only on the composition, density, and temperature. Although a variety of different schemes have been developed to compute a wavelength-independent opacity, by far the most commonly used is the **Rosseland mean opacity**, often simply referred to as the **Rosseland mean**.¹⁵ This *harmonic mean* gives the greatest contribution to the lowest values of opacity. In addition, the Rosseland mean incorporates a weighting function that depends on the rate at which the blackbody spectrum varies with temperature (recall Eq. 3.24).¹⁶ Formally, the Rosseland mean opacity is defined as

$$\frac{1}{\bar{\kappa}} \equiv \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu(T)}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu(T)}{\partial T} d\nu}. \quad (9.21)$$

Unfortunately, there is no simple equation that is capable of describing all of the complex contributions to the opacity by individual spectral lines in bound–bound transitions, and so an analytic expression for the Rosseland mean cannot be given for these processes. However, approximation formulae have been developed for both the average bound–free and free–free opacities:

$$\bar{\kappa}_{\text{bf}} = 4.34 \times 10^{21} \frac{g_{\text{bf}}}{t} Z(1 + X) \frac{\rho}{T^{3.5}} \text{ m}^2 \text{ kg}^{-1} \quad (9.22)$$

$$\bar{\kappa}_{\text{ff}} = 3.68 \times 10^{18} g_{\text{ff}} (1 - Z)(1 + X) \frac{\rho}{T^{3.5}} \text{ m}^2 \text{ kg}^{-1}, \quad (9.23)$$

where ρ is the density (in kg m^{-3}) and T is the temperature (in kelvins). X and Z are the **mass fractions**, or fractional abundances (by mass), of hydrogen and metals, respectively.¹⁷

¹⁴The additional dependencies of the opacity on the electron number density, states of excitation and ionization of the atoms and ions, and other factors can all be calculated from the composition, density, and temperature.

¹⁵This wavelength-averaged opacity was introduced in 1924 by the Norwegian astronomer Svein Rosseland (1894–1985).

¹⁶You may also wish to refer to Problem 7.2 for a discussion of the role of weighting functions in integral averages.

¹⁷As we noted on page 204, because the primary components of most stellar gases are hydrogen and helium, all other constituents are frequently lumped together and referred to as *metals*. In certain applications, however, it is necessary to specify the composition in greater detail. In these cases, each species is represented by its own mass fraction.

Together with the mass fraction of helium, Y , their formal definitions are

$$X \equiv \frac{\text{total mass of hydrogen}}{\text{total mass of gas}} \quad (9.24)$$

$$Y \equiv \frac{\text{total mass of helium}}{\text{total mass of gas}} \quad (9.25)$$

$$Z \equiv \frac{\text{total mass of metals}}{\text{total mass of gas}}. \quad (9.26)$$

Clearly, $X + Y + Z = 1$.

The **Gaunt factors**, g_{bf} and g_{ff} , are quantum-mechanical correction terms first calculated by J. A. Gaunt. These Gaunt factors are both ≈ 1 for the visible and ultraviolet wavelengths of interest in stellar atmospheres. The additional correction factor, t , in the equation for the bound-free opacity is called the **guillotine factor** and describes the cutoff of an atom's contribution to the opacity after it has been ionized. Typical values of t lie between 1 and 100.

Both of these formulae have the functional form $\bar{\kappa} = \kappa_0 \rho / T^{3.5}$, where κ_0 is approximately constant for a given composition. The first forms of these expressions were derived by H. A. Kramers (1894–1952) in 1923 using classical physics and the Rosseland mean. Any opacity having this density and temperature dependence is referred to as a **Kramers opacity law**.

Because the cross section for electron scattering is independent of wavelength, the Rosseland mean for this case has the particularly simple form

$$\bar{\kappa}_{\text{es}} = 0.02(1 + X) \text{ m}^2 \text{ kg}^{-1}. \quad (9.27)$$

An estimate of the contribution to the mean opacity provided by the H^- ion may also be included over the temperature range $3000 \text{ K} \leq T \leq 6000 \text{ K}$ and for densities between $10^{-7} \text{ kg m}^{-3} \leq \rho \leq 10^{-2} \text{ kg m}^{-3}$ when $X \sim 0.7$ and $0.001 < Z < 0.03$ (the values of X and Z are typical of main-sequence stars). Specifically,

$$\bar{\kappa}_{\text{H}^-} \approx 7.9 \times 10^{-34} (Z/0.02) \rho^{1/2} T^9 \text{ m}^2 \text{ kg}^{-1}. \quad (9.28)$$

The total Rosseland mean opacity, $\bar{\kappa}$, is the average of the sum of the individual contributors to the opacity:

$$\bar{\kappa} = \frac{\kappa_{\text{bb}} + \kappa_{\text{bf}} + \kappa_{\text{ff}} + \kappa_{\text{es}} + \kappa_{\text{H}^-}}{5}.$$

Figure 9.10 shows the results of an extensive computer calculation of the Rosseland mean opacity from first principles using detailed quantum physics. The calculation was carried out by Carlos Iglesias and Forrest Rogers for a composition with $X = 0.70$ and $Z = 0.02$.¹⁸ The values of $\bar{\kappa}$ are plotted as a function of the temperature for several densities.

¹⁸A specific mixture of elements known as the Anders–Grevesse abundances were used to calculate the opacities shown.

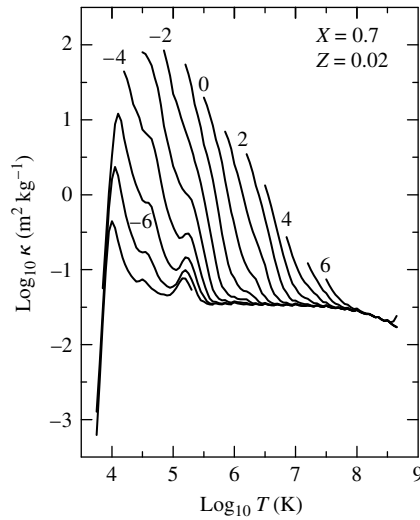


FIGURE 9.10 Rosseland mean opacity for a composition that is 70% hydrogen, 28% helium, and 2% metals by mass. The curves are labeled by the logarithmic value of the density ($\log_{10} \rho$ in kg m^{-3}). (Data from Iglesias and Rogers, *Ap. J.*, 464, 943, 1996.)

Considering the details of Fig. 9.10, first notice that the opacity increases with increasing density for a given temperature. Next, starting at the left-hand side of the figure, follow a constant-density plot as it rises steeply with increasing temperature. This reflects the increase in the number of free electrons produced by the ionization of hydrogen and helium. (Recall from Example 8.1.4 that the hydrogen partial ionization zone has a characteristic temperature of 10,000 K, and neutral helium is ionized at about the same temperature.) The decline of the plot after the peak in the opacity roughly follows a Kramers law, $\bar{\kappa} \propto T^{-3.5}$, and is due primarily to the bound–free and free–free absorption of photons. The He II ion loses its remaining electron at a characteristic temperature of 40,000 K for a wide range of stellar parameters; the slight increase in the number of free electrons produces a small “bump” seen near that temperature. Another bump, evident above 10^5 K, is the result of the ionization of certain metals, most notably iron. Finally, the plot reaches a flat floor at the right-hand side of the figure. Electron scattering dominates at the highest temperatures, when nearly all of the stellar material is ionized and there are few bound electrons available for bound–bound and bound–free processes. The form of Eq. (9.27) for electron scattering, with no density or temperature dependence, requires that all of the curves in Fig. 9.10 converge to the same constant value in the high-temperature limit.

9.3 ■ RADIATIVE TRANSFER

In an equilibrium, steady-state star, there can be no change in the total energy contained within any layer of the stellar atmosphere or interior.¹⁹ In other words, the mechanisms

¹⁹This is not the case for a star that is *not* in equilibrium. For instance, pulsating stars (to be discussed in Chapter 14), periodically absorb or “dam up” the outward flow of energy, driving the oscillations.

involved in absorbing and emitting energy must be precisely in balance throughout the star. In this section, the competition between the absorption and emission processes will be described, first in qualitative terms and later in more quantitative detail.

Photon Emission Processes

Any process that adds photons to a beam of light will be called **emission**. Thus emission includes the scattering of photons into the beam, as well as the true emission of photons by electrons making downward atomic transitions. Each of the four primary sources of opacity listed in Section 9.2 has an inverse emission process: bound–bound and free–bound emission, free–free emission (bremsstrahlung), and electron scattering. The simultaneous and complementary processes of absorption and emission hinder the flow of photons through the star by redirecting the paths of the photons and redistributing their energy. Thus in a star there is not a direct flow of photons streaming toward the surface, carrying energy outward at the speed of light. Instead, the individual photons travel only temporarily with the beam as they are repeatedly scattered in random directions following their encounters with gas particles.

The Random Walk

As the photons diffuse upward through the stellar material, they follow a haphazard path called a **random walk**. Figure 9.11 shows a photon that undergoes a net vector displacement $\mathbf{d}$ as the result of making a large number N of randomly directed steps, each of length ℓ (the mean free path):

$$\mathbf{d} = \ell_1 + \ell_2 + \ell_3 + \cdots + \ell_N.$$

Taking the vector dot product of $\mathbf{d}$ with itself gives

$$\begin{aligned} \mathbf{d} \cdot \mathbf{d} &= \ell_1 \cdot \ell_1 + \ell_1 \cdot \ell_2 + \cdots + \ell_1 \cdot \ell_N \\ &\quad + \ell_2 \cdot \ell_1 + \ell_2 \cdot \ell_2 + \cdots + \ell_2 \cdot \ell_N \\ &\quad + \cdots + \ell_N \cdot \ell_1 + \ell_N \cdot \ell_2 + \cdots + \ell_N \cdot \ell_N \\ &= \sum_{i=1}^N \sum_{j=1}^N \ell_i \cdot \ell_j, \end{aligned}$$

or

$$\begin{aligned} d^2 &= N\ell^2 + \ell^2 [\cos \theta_{12} + \cos \theta_{13} + \cdots + \cos \theta_{1N} \\ &\quad + \cos \theta_{21} + \cos \theta_{23} + \cdots + \cos \theta_{2N} \\ &\quad + \cdots + \cos \theta_{N1} + \cos \theta_{N2} + \cdots + \cos \theta_{N(N-1)}] \\ &= N\ell^2 + \ell^2 \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \cos \theta_{ij}, \end{aligned}$$

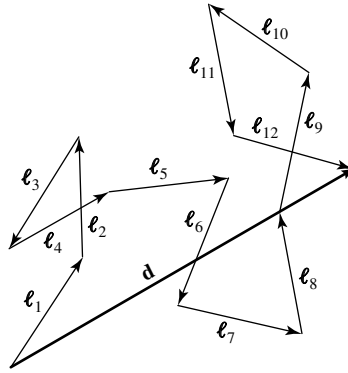


FIGURE 9.11 Displacement $\mathbf{d}$ of a random-walking photon.

where θ_{ij} is the angle between the vectors ℓ_i and ℓ_j . For a large number of randomly directed steps, the sum of all the cosine terms approaches zero. As a result, for a random walk, the displacement d is related to the size of each step, ℓ , by

$$d = \ell\sqrt{N}. \quad (9.29)$$

Thus the transport of energy through a star by radiation may be extremely inefficient. As a photon follows its tortuous path to the surface of a star,²⁰ it takes 100 steps to travel a distance of 10ℓ ; 10,000 steps to travel 100ℓ ; and one million steps to travel 1000ℓ .²¹ Because the optical depth at a point is roughly the number of photon mean free paths from that point to the surface (as measured along a light ray's straight path), Eq. (9.29) implies that the distance to the surface is $d = \tau_\lambda \ell = \ell\sqrt{N}$. The average number of steps needed for a photon to travel the distance d before leaving the surface is then

$$N = \tau_\lambda^2, \quad (9.30)$$

for $\tau_\lambda \gg 1$. As might be expected, when $\tau_\lambda \approx 1$, a photon may escape from that level of the star. A more careful analysis (performed in Section 9.4) shows that the average level in the atmosphere from which photons of wavelength λ escape is at a characteristic optical depth of about $\tau_\lambda = 2/3$. *Looking into a star at any angle, we always look back to an optical depth of about $\tau_\lambda = 2/3$, as measured straight back along the line of sight.* In fact, a star's photosphere is defined as the layer from which its visible light originates—that is, where $\tau_\lambda \approx 2/3$ for wavelengths in the star's continuum.

The realization that an observer looking vertically down on the surface of a star sees photons from $\tau_\lambda \approx 2/3$ offers an important insight into the formation of spectral lines.

²⁰Strictly speaking, an individual photon does not make the entire journey, but rather, along with being scattered, photons may be absorbed and re-emitted during the “collisions.”

²¹As will be discussed in Section 10.4, the process of transporting energy by radiation is sometimes so inefficient that another transport process, *convection*, must take over.

Recalling the definition of optical depth, Eq. (9.17),

$$\tau_\lambda = \int_0^s \kappa_\lambda \rho \, ds,$$

we see that if the opacity κ_λ increases at some wavelength, then the actual distance back along the ray to the level where $\tau_\lambda = 2/3$ decreases for that wavelength. One cannot see as far into murky material, so an observer will not see as deeply into the star at wavelengths where the opacity is greater than average (i.e., greater than the continuum opacity). This implies that if the temperature of the stellar atmosphere decreases outward, then these higher regions of the atmosphere will be cooler. As a result, the intensity of the radiation at $\tau_\lambda \approx 2/3$ will decline the most for those wavelengths at which the opacity is greatest, resulting in absorption lines in the continuous spectrum. Therefore, the temperature *must* decrease outward for the formation of absorption lines. This is the analog for stellar atmospheres of Kirchhoff's law that a cool, diffuse gas in front of a source of a continuous spectrum produces dark spectral lines in the continuous spectrum.

Limb Darkening

Another implication of receiving radiation from an optical depth of about two-thirds is shown in Fig. 9.12. The line of sight of an observer on Earth viewing the Sun is vertically downward at the center of the Sun's disk but makes an increasingly larger angle θ with the vertical near the edge, or *limb*, of the Sun. Looking near the limb, the observer will not see as deeply into the solar atmosphere and will therefore see a lower temperature at an optical depth of two-thirds (compared to looking at the center of the disk). As a result, the limb of

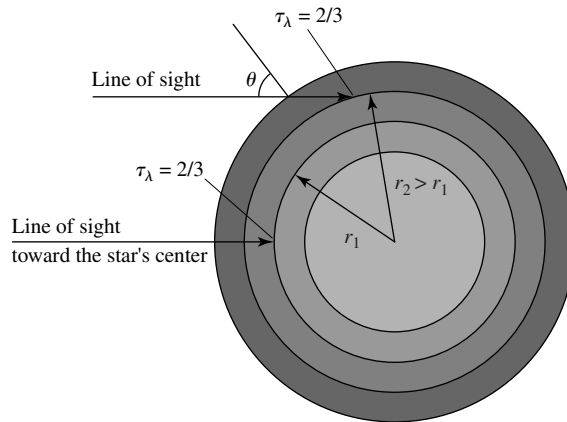


FIGURE 9.12 Limb darkening. The distance traversed within the atmosphere of the star to reach a specified radial distance r from the star's center increases along the line of sight of the observer as θ increases. This implies that to reach a specified optical depth (e.g., $\tau_\lambda = 2/3$), the line of sight terminates at greater distances (and cooler temperatures) from the star's center as θ increases. Note that the physical scale of the photosphere has been greatly exaggerated for illustration purposes. The thickness of a typical photosphere is on the order of 0.1% of the stellar radius.

the Sun appears darker than its center. This **limb darkening** is clearly seen in Fig. 11.11 for the Sun and has also been observed in the light curves of some eclipsing binaries. More detailed information on limb darkening may be found later in this section.

The Radiation Pressure Gradient

Considering the meandering nature of a photon's journey to the surface, it may seem surprising that the energy from the deep interior of the star ever manages to escape into space. At great depth in the interior of the star, the photon's mean free path is only a fraction of a centimeter. After a few scattering encounters, the photon is traveling in a nearly random direction, hundreds of millions of meters from the surface. This situation is analogous to the motions of air molecules in a closed room. An individual molecule moves about with a speed of nearly 500 m s^{-1} , and it collides with other air molecules several billion times per second. As a result, the molecules are moving in random directions. Because there is no overall migration of the molecules in a closed room, a person standing in the room feels no wind. However, opening a window may cause a breeze if a pressure difference is established between one side of the room and the other. The air in the room responds to this pressure gradient, producing a net flux of molecules toward the area of lower pressure.

In a star the same mechanism causes a “breeze” of photons to move toward the surface of the star. Because the temperature in a star decreases outward, the radiation pressure is smaller at greater distances from the center (cf., Eq. 9.11 for the blackbody radiation pressure). This gradient in the radiation pressure produces the slight net movement of photons toward the surface that carries the radiative flux. As we will discover later in this section, this process is described by

$$\frac{dP_{\text{rad}}}{dr} = -\frac{\bar{\kappa}\rho}{c} F_{\text{rad}}. \quad (9.31)$$

Thus the transfer of energy by radiation is a subtle process involving the slow upward diffusion of randomly walking photons, drifting toward the surface in response to minute differences in the radiation pressure. The description of a “beam” or a “ray” of light is only a convenient fiction, used to define the direction of motion momentarily shared by the photons that are continually absorbed and scattered into and out of the beam. Nevertheless, we will continue to use the language of photons traveling in a beam or ray of light, realizing that a specific photon is in the beam for only an instant.

9.4 ■ THE TRANSFER EQUATION

In this section, we will focus on a more thorough examination of the flow of radiation through a stellar atmosphere.²² We will develop and solve the basic equation of radiative transfer using several standard assumptions. In addition, we will derive the variation of temperature with optical depth in a simple model atmosphere before applying it to obtain a quantitative description of limb darkening.

²²Although the focus of this discussion is on stellar atmospheres, much of the discussion is applicable to other environments as well, such as light traversing an interstellar gas cloud.

The Emission Coefficient

In the following discussions of beams and light rays, the primary consideration is the net flow of energy in a given direction, not the specific path taken by individual photons. First, we will examine the emission process that increases the intensity of a ray of wavelength λ as it travels through a gas. The increase in intensity dI_λ is proportional to both ds , the distance traveled in the direction of the ray, and ρ , the density of the gas. For pure emission (no absorption of the radiation),

$$dI_\lambda = j_\lambda \rho ds, \quad (9.32)$$

where j_λ is the **emission coefficient** of the gas. The emission coefficient, which has units of $\text{m s}^{-3} \text{sr}^{-1}$, varies with the wavelength of the light.

As a beam of light moves through the gas in a star, its specific intensity, I_λ , changes as photons traveling with the beam are removed by absorption or scattering out of the beam, and are replaced by photons emitted from the surrounding stellar material, or scattered into the beam. Combining Eq. (9.13) for the decrease in intensity due to the absorption of radiation with Eq. (9.32) for the increase produced by emission gives the general result

$$dI_\lambda = -\kappa_\lambda \rho I_\lambda ds + j_\lambda \rho ds. \quad (9.33)$$

The ratio of the rates at which the competing processes of emission and absorption occur determines how rapidly the intensity of the beam changes. This is similar to describing the flow of traffic on an interstate highway. Imagine following a group of cars as they leave Los Angeles, traveling north on I-15. Initially, nearly all of the cars on the road have California license plates. Driving north, the number of cars on the road declines as more individuals exit than enter the highway. Eventually approaching Las Vegas, the number of cars on the road increases again, but now the surrounding cars bear Nevada license plates. Continuing onward, the traffic fluctuates as the license plates eventually change to those of Utah, Idaho, and Montana. Most of the cars have the plates of the state they are in, with a few cars from neighboring states and even fewer from more distant locales. At any point along the way, the number of cars on the road reflects the local population density. Of course, this is to be expected; the surrounding area is the source of the traffic entering the highway, and the rate at which the traffic changes is determined by the ratio of the number of entering to exiting automobiles. This ratio determines how rapidly the cars on the road from elsewhere are replaced by the cars belonging to the local population. Thus the traffic constantly changes, always tending to resemble the number and types of automobiles driven by the people living nearby.

The Source Function and the Transfer Equation

In a stellar atmosphere or interior, the same considerations describe the competition between the rates at which photons are plucked out of a beam of light by absorption, and introduced into the beam by emission processes. The ratio of the rates of emission and absorption determines how rapidly the intensity of the beam of light changes and describes the tendency of the population of photons in the beam to resemble the local source of photons in the surrounding stellar material. To introduce the ratio of emission to absorption, we divide

Eq. (9.33) by $-\kappa_\lambda \rho ds$:

$$-\frac{1}{\kappa_\lambda \rho} \frac{dI_\lambda}{ds} = I_\lambda - \frac{j_\lambda}{\kappa_\lambda}.$$

The ratio of the emission coefficient to the absorption coefficient is called the **source function**, $S_\lambda \equiv j_\lambda/\kappa_\lambda$. It describes how photons originally traveling with the beam are removed and replaced by photons from the surrounding gas.²³ The source function, S_λ , has the same units as the intensity, $\text{W m}^{-3} \text{sr}^{-1}$. Therefore, in terms of the source function,

$$\boxed{-\frac{1}{\kappa_\lambda \rho} \frac{dI_\lambda}{ds} = I_\lambda - S_\lambda.} \quad (9.34)$$

This is one form of the **equation of radiative transfer** (usually referred to as the **transfer equation**).²⁴ According to the transfer equation, if the intensity of the light does not vary (so that the left-hand side of the equation is zero), then the intensity is equal to the source function, $I_\lambda = S_\lambda$. If the intensity of the light is *greater* than the source function (the right-hand side of the transfer equation is greater than 0), then dI_λ/ds is less than 0, and the intensity *decreases* with distance. On the other hand, if the intensity is *less* than the source function, the intensity *increases* with distance. This is merely a mathematical restatement of the tendency of the photons found in the beam to resemble the local source of photons in the surrounding gas. Thus *the intensity of the light tends to become equal to the local value of the source function*, although the source function itself may vary too rapidly with distance for an equality to be attained.

The Special Case of Blackbody Radiation

The source function for the special case of blackbody radiation can be found by considering a box of optically thick gas maintained at a constant temperature T . The confined particles and blackbody radiation are in thermodynamic equilibrium, with no net flow of energy through the box or between the gas particles and the radiation. With the particles and photons in equilibrium, individually and with each other, every process of absorption is balanced by an inverse process of emission. The intensity of the radiation is described by the Planck function, $I_\lambda = B_\lambda$. Furthermore, because the intensity is constant throughout the box, $dI_\lambda/ds = 0$, and so $I_\lambda = S_\lambda$. *For the case of thermodynamic equilibrium, the source function is equal to the Planck function, $S_\lambda = B_\lambda$.*

As mentioned in Section 9.2, a star cannot be in perfect thermodynamic equilibrium; there is a net flow of energy from the center to the surface. Deep in the atmosphere, where $\tau_\lambda \gg 1$ as measured along a vertical ray, a random-walking photon will take at least τ_λ^2 steps to reach the surface (recall Eq. 9.30) and so will suffer many scattering events before escaping from the star. Thus, at a depth at which the photon mean free path is small compared to the

²³As a ratio involving the inverse processes of absorption and emission, the source function is less sensitive to the detailed properties of the stellar material than are j_λ and κ_λ individually.

²⁴It is assumed that the atmosphere is in a steady state, not changing with time. Otherwise, a time-derivative term would have to be included in the transfer equation.

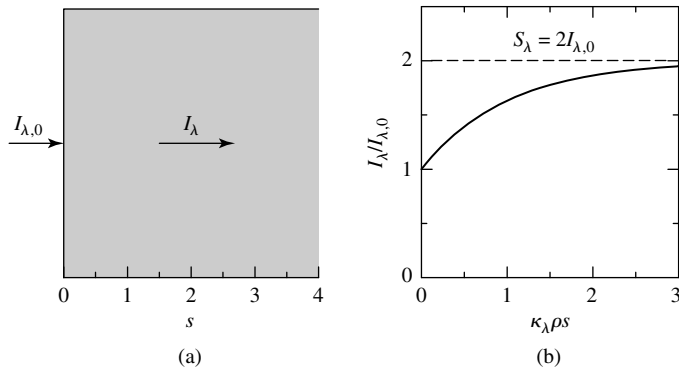


FIGURE 9.13 Transformation of the intensity of a light ray traveling through a volume of gas. (a) A light ray entering a volume of gas. (b) Intensity of the light ray. The horizontal axis has units of $\kappa_{\lambda}\rho s$, the number of optical depths traveled into the gas.

temperature scale height, the photons are effectively confined to a limited volume, a region of nearly constant temperature. The conditions for local thermodynamic equilibrium (LTE) are satisfied, and so, as already seen, the source function is equal to the Planck function, $S_{\lambda} = B_{\lambda}$. Making the assumption of LTE in a problem means setting $S_{\lambda} = B_{\lambda}$. However, even in LTE, the intensity of the radiation, I_{λ} , will not necessarily be equal to B_{λ} unless $\tau_{\lambda} \gg 1$. In summary, saying that $I_{\lambda} = B_{\lambda}$ is a statement that the radiation field is described by the Planck function, while $S_{\lambda} = B_{\lambda}$ describes the physical source of the radiation, $j_{\lambda}/\kappa_{\lambda}$, as one that produces blackbody radiation.

Example 9.4.1. To see how the intensity of a light ray tends to become equal to the local value of the source function, imagine a beam of light of initial intensity $I_{\lambda,0}$ at $s = 0$ entering a volume of gas of constant density, ρ , that has a *constant* opacity, κ_{λ} , and a *constant* source function, S_{λ} . Then it is left as an exercise to show that the transfer equation (Eq. 9.34) may be easily solved for the intensity of the light as a function of the distance s traveled into the gas:

$$I_{\lambda}(s) = I_{\lambda,0} e^{-\kappa_{\lambda}\rho s} + S_{\lambda}(1 - e^{-\kappa_{\lambda}\rho s}). \quad (9.35)$$

As shown in Fig. 9.13 for the case of $S_{\lambda} = 2I_{\lambda,0}$, this solution describes the transformation of the intensity of the light ray from its initial value of $I_{\lambda,0}$ to S_{λ} , the value of the source function. The characteristic distance for this change to occur is $s = 1/\kappa_{\lambda}\rho$, which is one photon mean free path (recall Example 9.2.2), or one optical depth into the gas.

The Assumption of a Plane-Parallel Atmosphere

Although the transfer equation is the basic tool that describes the passage of light through a star's atmosphere, a reader seeing it for the first time may be prone to despair. In this troublesome equation, the intensity of the light must depend on the direction of travel to account

for the net outward flow of energy. And although absorption and emission coefficients are the same for light traveling in all directions (implying that the source function is independent of direction), the absorption and emission coefficients depend on the temperature and density in a rather complicated way.

However, if astronomers are to learn anything about the physical conditions in stellar atmospheres, such as temperature or density, they must know where (at what depth) a spectral line is formed. A vast amount of effort has therefore been devoted to solving and understanding the implications of the transfer equation, and several powerful techniques have been developed that simplify the analysis considerably.

We will begin by rewriting Eq. (9.34) in terms of the optical depth τ_λ , defined by Eq. (9.15), resulting in

$$\frac{dI_\lambda}{d\tau_\lambda} = I_\lambda - S_\lambda. \quad (9.36)$$

Unfortunately, because the optical depth is measured along the path of the light ray, neither the optical depth nor the distance s in Eq. (9.34) corresponds to a unique geometric depth in the atmosphere. Consequently, the optical depth must be replaced by a meaningful measure of position.

To find a suitable replacement, we introduce the first of several standard approximations. The atmospheres of stars near the main sequence are physically thin compared with the size of the star, analogous to the skin of an onion. The atmosphere's radius of curvature is thus much larger than its thickness, and we may consider the atmosphere as a *plane-parallel slab*. As shown in Fig. 9.14, the z -axis is assumed to be in the vertical direction, with $z = 0$ at the top of this **plane-parallel atmosphere**.

Next, a **vertical optical depth**, $\tau_{\lambda,v}(z)$, is defined as

$$\tau_{\lambda,v}(z) \equiv \int_z^0 \kappa_\lambda \rho \, dz. \quad (9.37)$$

Comparison with Eq. (9.17) reveals that this is just the initial optical depth of a ray traveling

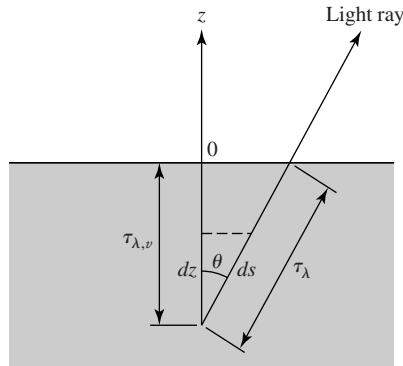


FIGURE 9.14 Plane-parallel stellar atmosphere.

vertically upward from an initial position ($z < 0$) to the surface ($z = 0$) where $\tau_{\lambda,v} = 0$.²⁵ However, a ray that travels upward at an angle θ from the same initial position z has farther to go through the same layers of the atmosphere in order to reach the surface. Therefore, the optical depth measured along this ray's path to the surface, τ_λ , is *greater* than the vertical optical depth, $\tau_{\lambda,v}(z)$. Since $dz = ds \cos \theta$, the two optical depths are related by

$$\tau_\lambda = \frac{\tau_{\lambda,v}}{\cos \theta} = \tau_{\lambda,v} \sec \theta. \quad (9.38)$$

The vertical optical depth is a true vertical coordinate, analogous to z , that increases in the $-z$ -direction. Its value does not depend on the direction of travel of a light ray, and so it can be used as a meaningful position coordinate in the transfer equation. Replacing τ_λ by $\tau_{\lambda,v}$ in Eq. (9.36) results in

$$\cos \theta \frac{dI_\lambda}{d\tau_{\lambda,v}} = I_\lambda - S_\lambda. \quad (9.39)$$

This form of the transfer equation is usually employed when dealing with the approximation of a plane-parallel atmosphere.

Of course, the value of the vertical optical depth at a level z is wavelength-dependent because of the wavelength-dependent opacity in Eq. (9.37). In order to simplify the following analysis, and to permit the identification of an atmospheric level with a unique value of τ_v , the opacity is assumed to be *independent of wavelength* (we usually take it to be equal to the Rosseland mean opacity, $\bar{\kappa}$). A model stellar atmosphere, for which the simplifying assumption is made that the opacity is independent of wavelength, is called a *gray atmosphere*, reflecting its indifference to the spectrum of wavelengths. If we write $\bar{\kappa}$ instead of κ_λ in Eq. (9.37), the vertical optical depth no longer depends on wavelength; we can therefore write τ_v instead of $\tau_{\lambda,v}$ in the transfer equation (Eq. 9.39). The remaining wavelength dependencies may be removed by integrating the transfer equation over all wavelengths, using

$$I = \int_0^\infty I_\lambda d\lambda \quad \text{and} \quad S = \int_0^\infty S_\lambda d\lambda.$$

With the preceding changes, the transfer equation appropriate for a plane-parallel gray atmosphere is

$$\cos \theta \frac{dI}{d\tau_v} = I - S. \quad (9.40)$$

This equation leads to two particularly useful relations between the various quantities describing the radiation field. First, integrating over all solid angles, and recalling that S depends only on the local conditions of the gas, independent of direction, we get

$$\frac{d}{d\tau_v} \int I \cos \theta d\Omega = \int I d\Omega - S \int d\Omega. \quad (9.41)$$

²⁵Recall that as the light approaches the surface (and the observer on Earth), it is traveling through smaller and smaller values of the optical depth.

Using $\int d\Omega = 4\pi$ together with the definitions of the radiative flux F_{rad} (Eq. 9.8) and the mean intensity $\langle I \rangle$ (Eq. 9.3), both integrated over all wavelengths, we find

$$\frac{dF_{\text{rad}}}{d\tau_v} = 4\pi(\langle I \rangle - S).$$

The second relation is found by first multiplying the transfer equation (9.40) by $\cos \theta$ and again integrating over all solid angles:

$$\frac{d}{d\tau_v} \int I \cos^2 \theta d\Omega = \int I \cos \theta d\Omega - S \int \cos \theta d\Omega.$$

The term on the left is the radiation pressure multiplied by the speed of light (recall Eq. 9.9). The first term on the right-hand side is the radiative flux. In spherical coordinates, the second integral on the right-hand side evaluates to

$$\int \cos \theta d\Omega = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \cos \theta \sin \theta d\theta d\phi = 0.$$

Thus

$$\frac{dP_{\text{rad}}}{d\tau_v} = \frac{1}{c} F_{\text{rad}}. \quad (9.42)$$

In Problem 9.16, you will find that in a spherical coordinate system with its origin at the center of the star, this equation is

$$\frac{dP_{\text{rad}}}{dr} = -\frac{\bar{\kappa}\rho}{c} F_{\text{rad}},$$

which is just Eq. (9.31). As mentioned previously, this result can be interpreted as saying that the net radiative flux is driven by differences in the radiation pressure, with a “photon wind” blowing from high to low P_{rad} . Equation (9.31) will be employed in Chapter 10 to determine the temperature structure in the interior of a star.

In an equilibrium stellar atmosphere, every process of absorption is balanced by an inverse process of emission; no net energy is subtracted from or added to the radiation field. In a plane-parallel atmosphere, this means that the radiative flux must have the same value at every level of the atmosphere, including its surface. From Eq. (3.18),

$$F_{\text{rad}} = \text{constant} = F_{\text{surf}} = \sigma T_e^4. \quad (9.43)$$

Because the flux is a constant, $dF_{\text{rad}}/d\tau_v = 0$, which implies that the mean intensity must be equal to the source function,

$$\langle I \rangle = S. \quad (9.44)$$

Equation (9.42) may now be integrated to find the radiation pressure as a function of the vertical optical depth:

$$P_{\text{rad}} = \frac{1}{c} F_{\text{rad}} \tau_v + C, \quad (9.45)$$

where C is the constant of integration.

The Eddington Approximation

If we knew how the radiation pressure varied with temperature for the general case (and not just for blackbody radiation), we could use Eq. (9.45) to determine the temperature structure of our plane-parallel gray atmosphere. We would have to *assume* a description of the angular distribution of the intensity. In an approximation that we owe to the brilliant English physicist Sir Arthur Stanley Eddington (1882–1944), the intensity of the radiation field is assigned one value, I_{out} , in the $+z$ -direction (outward) and another value, I_{in} , in the $-z$ -direction (inward); see Fig. 9.15. Both I_{out} and I_{in} vary with depth in the atmosphere, and in particular, $I_{\text{in}} = 0$ at the top of the atmosphere, where $\tau_v = 0$. It is left as an exercise to show that with this **Eddington approximation**,²⁶ the mean intensity, radiative flux, and radiation pressure are given by

$$\langle I \rangle = \frac{1}{2} (I_{\text{out}} + I_{\text{in}}) \quad (9.46)$$

$$F_{\text{rad}} = \pi (I_{\text{out}} - I_{\text{in}}) \quad (9.47)$$

$$P_{\text{rad}} = \frac{2\pi}{3c} (I_{\text{out}} + I_{\text{in}}) = \frac{4\pi}{3c} \langle I \rangle. \quad (9.48)$$

[Note that because the flux is a constant, Eq. (9.47) shows that there is a constant difference between I_{out} and I_{in} at any level of the atmosphere.]

Inserting the last relation for the radiation pressure into Eq. (9.45), we find that

$$\frac{4\pi}{3c} \langle I \rangle = \frac{1}{c} F_{\text{rad}} \tau_v + C. \quad (9.49)$$

The constant C can be determined by evaluating Eqs. (9.46) and (9.47) at the top of the atmosphere, where $\tau_v = 0$ and $I_{\text{in}} = 0$. The result is that $\langle I(\tau_v = 0) \rangle = F_{\text{rad}}/2\pi$. Inserting

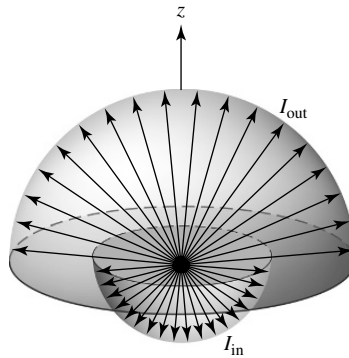


FIGURE 9.15 The Eddington approximation.

²⁶Actually, there are several more mathematical ways of implementing the Eddington approximation, but they are all equivalent.

this into Eq. (9.49) with $\tau_v = 0$ shows that

$$C = \frac{2}{3c} F_{\text{rad}}.$$

With this value of C , Eq. (9.49) becomes

$$\frac{4\pi}{3} \langle I \rangle = F_{\text{rad}} \left(\tau_v + \frac{2}{3} \right). \quad (9.50)$$

Of course, we already know that the radiative flux is a constant, given by Eq. (9.43). Using this results in an expression for the mean intensity as a function of the vertical optical depth:

$$\langle I \rangle = \frac{3\sigma}{4\pi} T_e^4 \left(\tau_v + \frac{2}{3} \right). \quad (9.51)$$

We may now derive the final approximation to determine the temperature structure of our model atmosphere. If the atmosphere is assumed to be in local thermodynamic equilibrium, another expression for the mean intensity can be found and combined with Eq. (9.51). By the definition of LTE, the source function is equal to the Planck function, $S_\lambda = B_\lambda$. Integrating B_λ over all wavelengths (see Eq. 3.28) shows that for LTE,

$$S = B = \frac{\sigma T^4}{\pi},$$

and so, from Eq. (9.44),

$$\langle I \rangle = \frac{\sigma T^4}{\pi}. \quad (9.52)$$

Equating expressions (9.51) and (9.52) finally results in the variation of the temperature with vertical optical depth in a plane-parallel gray atmosphere in LTE, assuming the Eddington approximation:²⁷

$$T^4 = \frac{3}{4} T_e^4 \left(\tau_v + \frac{2}{3} \right). \quad (9.53)$$

This relation is well worth the effort of its derivation, because it reveals some important aspects of real stellar atmospheres. First, notice that $T = T_e$ at $\tau_v = 2/3$, *not* at $\tau_v = 0$. Thus the “surface” of a star, which by definition has temperature T_e [recall the Stefan–Boltzmann equation, Eq. (3.17)], is *not* at the top of the atmosphere, where $\tau_v = 0$, but deeper down, where $\tau_v = 2/3$. This result may be thought of as the average point of origin of the observed photons. Although this result came at the end of a string of assumptions, it can be generalized to the statement that *when looking at a star, we see down to a vertical optical depth of $\tau_v \approx 2/3$, averaged over the disk of the star*. The importance of this for the formation and interpretation of spectral lines was discussed on page 254.

²⁷You are encouraged to refer to Mihalas, Chapter 3, for a more detailed discussion of the gray atmosphere, including a more sophisticated development of the relation $T^4 = \frac{3}{4} T_e^4 [\tau_v + q(\tau_v)]$, where the Eddington approximation [$q(\tau_v) \equiv \frac{2}{3}$] is a special case.

Limb Darkening Revisited

We now move on to take a closer look at limb darkening (recall Fig. 9.12). A comparison of theory and observations of limb darkening can provide valuable information about how the source function varies with depth in a star's atmosphere. To see how this is done, we first solve the general form of the transfer equation (Eq. 9.36),

$$\frac{dI_\lambda}{d\tau_\lambda} = I_\lambda - S_\lambda,$$

at least formally, rather than by making assumptions. (The inevitable assumptions will be required soon enough.) Multiplying both sides by $e^{-\tau_\lambda}$, we have

$$\frac{dI_\lambda}{d\tau_\lambda} e^{-\tau_\lambda} - I_\lambda e^{-\tau_\lambda} = -S_\lambda e^{-\tau_\lambda}$$

$$\frac{d}{d\tau_\lambda} (e^{-\tau_\lambda} I_\lambda) = -S_\lambda e^{-\tau_\lambda}$$

$$d(e^{-\tau_\lambda} I_\lambda) = -S_\lambda e^{-\tau_\lambda} d\tau_\lambda.$$

If we integrate from the initial position of the ray, at optical depth $\tau_{\lambda,0}$ where $I_\lambda = I_{\lambda,0}$, to the top of the atmosphere, at optical depth $\tau_\lambda = 0$ where $I_\lambda = I_\lambda(0)$, the result for the emergent intensity at the top of the atmosphere, $I_\lambda(0)$, is

$$I_\lambda(0) = I_{\lambda,0} e^{-\tau_{\lambda,0}} - \int_{\tau_{\lambda,0}}^0 S_\lambda e^{-\tau_\lambda} d\tau_\lambda. \quad (9.54)$$

This equation has a very straightforward interpretation. The emergent intensity on the left is the sum of two positive contributions. The first term on the right is the initial intensity of the ray, reduced by the effects of absorption along the path to the surface. The second term, also positive,²⁸ represents the emission at every point along the path, attenuated by the absorption between the point of emission and the surface.

We now return to the geometry of a plane-parallel atmosphere and the vertical optical depth, τ_v . However, we do *not* assume a gray atmosphere, LTE, or make the Eddington approximation. As shown in Fig. 9.16, the problem of limb darkening amounts to determining the emergent intensity $I_\lambda(0)$ as a function of the angle θ . Equation (9.54), the formal solution to the transfer equation, is easily converted to this situation by using Eq. (9.38) to replace τ_λ with $\tau_{\lambda,v} \sec \theta$ (the vertical optical depth) to get

$$I(0) = I_0 e^{-\tau_{v,0} \sec \theta} - \int_{\tau_{v,0} \sec \theta}^0 S \sec \theta e^{-\tau_v \sec \theta} d\tau_v.$$

Although both I and τ_v depend on wavelength, the λ subscript has been dropped to simplify the notation; the approximation of a gray atmosphere has *not* been made. To include the contributions to the emergent intensity from all layers of the atmosphere, we take the value

²⁸Remember that the optical depth, measured along the ray's path, decreases in the direction of travel, so $d\tau_\lambda$ is negative.

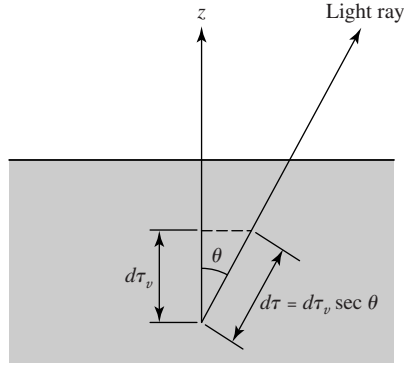


FIGURE 9.16 Finding $I(0)$ as a function of θ for limb darkening in plane-parallel geometry.

of the initial position of the rays to be at $\tau_{v,0} = \infty$. Then the first term on the right-hand side vanishes, leaving

$$I(0) = \int_0^\infty S \sec \theta e^{-\tau_v \sec \theta} d\tau_v. \quad (9.55)$$

If we knew how the source function depends on the vertical optical depth, this equation could be integrated to find the emergent intensity as a function of the direction of travel, θ , of the ray. Although the form of the source function is not known, a good guess will be enough to estimate $I(0)$. Suppose that the source function has the form

$$S = a + b\tau_v, \quad (9.56)$$

where a and b are wavelength-dependent numbers to be determined. Inserting this into Eq. (9.55) and integrating (the details are left as an exercise) show that the emergent intensity for this source function is

$$I_\lambda(0) = a_\lambda + b_\lambda \cos \theta, \quad (9.57)$$

where the λ subscripts have been restored to the appropriate quantities to emphasize their wavelength dependence. By making careful measurements of the variation in the specific intensity across the disk of the Sun, the values of a_λ and b_λ for the solar source function can be determined for a range of wavelengths. For example, for a wavelength of 501 nm, Böhm-Vitense (1989) supplies values of $a_{501} = 1.04 \times 10^{13} \text{ W m}^{-3} \text{ sr}^{-1}$ and $b_{501} = 3.52 \times 10^{13} \text{ W m}^{-3} \text{ sr}^{-1}$.

Example 9.4.2. Solar limb darkening provides an opportunity to test the accuracy of our “plane-parallel gray atmosphere in LTE using the Eddington approximation.” In the preceding discussion of an equilibrium gray atmosphere, it was found that the mean intensity is equal to the source function,

$$\langle I \rangle = S$$

continued

(Eq. 9.44). Then, with the additional assumptions of the Eddington approximation and LTE, Eqs. (9.52) and (9.53) can be used to determine the mean intensity and thus the source function:

$$S = \langle I \rangle = \frac{\sigma T^4}{\pi} = \frac{3\sigma}{4\pi} T_e^4 \left(\tau_v + \frac{2}{3} \right).$$

Taking the source function to have the form of Eq. (9.56), $S = a + b\tau_v$, as used earlier for limb darkening (*after integrating over all wavelengths*), the values of the coefficients are

$$a = \frac{\sigma}{2\pi} T_e^4 \quad \text{and} \quad b = \frac{3\sigma}{4\pi} T_e^4.$$

The emergent intensity then will have the form of Eq. (9.57), $I(0) = a + b \cos \theta$ (again after integrating over all wavelengths). The ratio of the emergent intensity at angle θ , $I(\theta)$, to that at the center of the star, $I(\theta = 0)$, is thus

$$\frac{I(\theta)}{I(\theta = 0)} = \frac{a + b \cos \theta}{a + b} = \frac{2}{5} + \frac{3}{5} \cos \theta. \quad (9.58)$$

We can compare the results of this calculation with observations of solar limb darkening in integrated light (made by summing over all wavelengths). Figure 9.17 shows both the observed values of $I(\theta)/I(\theta = 0)$ and the values from Eq. (9.58). The agreement is remarkably good, despite our numerous approximations. However, be forewarned that the agreement is much worse for observations made at a given wavelength (see Böhm-Vitense, 1989) as a consequence of wavelength-dependent opacity effects such as line blanketing.

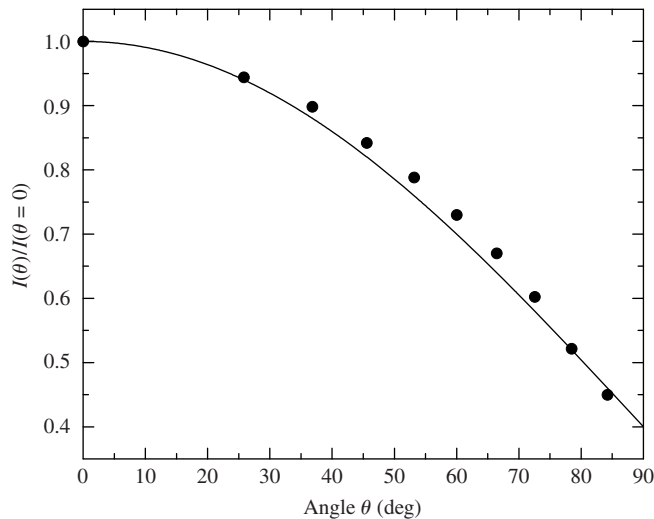


FIGURE 9.17 A theoretical Eddington approximation of solar limb darkening for light integrated over all wavelengths. The dots are observational data for the Sun. Although a good fit, the Eddington approximation is not perfect, which implies that a more detailed model must be developed; see, for example, Problem 9.29.

9.5 ■ THE PROFILES OF SPECTRAL LINES

We now have a formidable theoretical arsenal to bring to bear on the analysis of spectral lines. The shape of an individual spectral line contains a wealth of information about the environment in which it was formed.

Equivalent Widths

Figure 9.18 shows a graph of the radiant flux, F_λ , as a function of wavelength for a typical absorption line. In the figure, F_λ is expressed as a fraction of F_c , the value of the flux from the continuous spectrum outside the spectral line. Near the central wavelength, λ_0 , is the *core* of the line, and the sides sweeping upward to the continuum are the line's *wings*. Individual lines may be narrow or broad, shallow or deep. The quantity $(F_c - F_\lambda)/F_c$ is referred to as the *depth* of the line. The strength of a spectral line is measured in terms of its **equivalent width**. The equivalent width W of a spectral line is defined as the width of a box (shaded in Fig. 9.18) reaching up to the continuum that has the same area as the spectral line. That is,

$$W = \int \frac{F_c - F_\lambda}{F_c} d\lambda, \quad (9.59)$$

where the integral is taken from one side of the line to the other. The equivalent width of a line in the visible spectrum, shaded in Fig. 9.18, is usually on the order of 0.01 nm. Another measure of the width of a spectral line is the change in wavelength from one side of the line to the other, where its depth $(F_c - F_\lambda)/(F_c - F_{\lambda_0}) = 1/2$; this is called the *full width at half-maximum* and will be denoted by $(\Delta\lambda)_{1/2}$.

The spectral line shown in Fig. 9.18 is termed **optically thin** because there is no wavelength at which the radiant flux has been completely blocked. The opacity κ_λ of the stellar

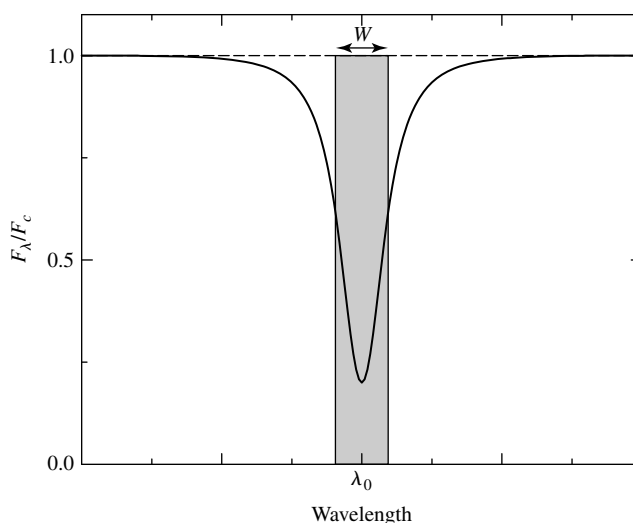


FIGURE 9.18 The profile of a typical spectral line.

material is greatest at the wavelength λ_0 at the line's center and decreases moving into the wings. From the discussion on page 254, this means that the center of the line is formed at higher (and cooler) regions of the stellar atmosphere. Moving into the wings from λ_0 , the line formation occurs at progressively deeper (and hotter) layers of the atmosphere, until it merges with the continuum-producing region at an optical depth of $2/3$. In Section 11.2 this idea will be applied to the absorption lines produced in the solar photosphere.

Processes That Broaden Spectral Lines

Three main processes are responsible for the broadening of spectral lines. Each of these mechanisms produces its own distinctive line shape or *line profile*.

1. **Natural broadening.** Spectral lines cannot be infinitely sharp, even for motionless, isolated atoms. According to Heisenberg's uncertainty principle (recall Eq. 5.20), as the time available for an energy measurement decreases, the inherent uncertainty of the result increases. Because an electron in an excited state occupies its orbital for only a brief instant, Δt , the orbital's energy, E , cannot have a precise value. Thus the uncertainty in the energy, ΔE , of the orbital is approximately

$$\Delta E \approx \frac{\hbar}{\Delta t}.$$

(The electron's lifetime in the ground state may be taken as infinite, so in that case $\Delta E = 0$.) Electrons can make transitions from and to anywhere within these "fuzzy" energy levels, producing an uncertainty in the wavelength of the photon absorbed or emitted in a transition. Using Eq. (5.3) for the energy of a photon, $E_{\text{photon}} = hc/\lambda$, we find that the uncertainty in the photon's wavelength has a magnitude of roughly

$$\Delta \lambda \approx \frac{\lambda^2}{2\pi c} \left(\frac{1}{\Delta t_i} + \frac{1}{\Delta t_f} \right), \quad (9.60)$$

where Δt_i is the lifetime of the electron in its initial state and Δt_f is the lifetime in the final state. (The proof is left as a problem.)

Example 9.5.1. The lifetime of an electron in the first and second excited states of hydrogen is about $\Delta t = 10^{-8}$ s. The natural broadening of the H α line of hydrogen, $\lambda = 656.3$ nm, is then

$$\Delta \lambda \approx 4.57 \times 10^{-14} \text{ m} = 4.57 \times 10^{-5} \text{ nm}.$$

A more involved calculation shows that the full width at half-maximum of the line profile for natural broadening is

$$(\Delta \lambda)_{1/2} = \frac{\lambda^2}{\pi c} \frac{1}{\Delta t_0}, \quad (9.61)$$

where Δt_0 is the average waiting time for a specific transition to occur. This results in a typical value of

$$(\Delta\lambda)_{1/2} \simeq 2.4 \times 10^{-5} \text{ nm},$$

in good agreement with the preceding estimate.

2. **Doppler broadening.** In thermal equilibrium, the atoms in a gas, each of mass m , are moving randomly about with a distribution of speeds that is described by the Maxwell–Boltzmann distribution function (Eq. 8.1), with the most probable speed given by Eq. (8.2), $v_{\text{mp}} = \sqrt{2kT/m}$. The wavelengths of the light absorbed or emitted by the atoms in the gas are Doppler-shifted according to (nonrelativistic) Eq. (4.30), $\Delta\lambda/\lambda = \pm |v_r|/c$. Thus the width of a spectral line due to Doppler broadening should be approximately

$$\Delta\lambda \approx \frac{2\lambda}{c} \sqrt{\frac{2kT}{m}}.$$

Example 9.5.2. For hydrogen atoms in the Sun’s photosphere ($T = 5777 \text{ K}$), the Doppler broadening of the $\text{H}\alpha$ line should be about

$$\Delta\lambda \approx 0.0427 \text{ nm},$$

roughly 1000 times greater than for natural broadening.

A more in-depth analysis, taking into account the different directions of the atoms’ motions with respect to one another and to the line of sight of the observer, shows that the full width at half-maximum of the line profile for Doppler broadening is

$$(\Delta\lambda)_{1/2} = \frac{2\lambda}{c} \sqrt{\frac{2kT \ln 2}{m}}. \quad (9.62)$$

Although the line profile for Doppler broadening is much wider at half-maximum than for natural broadening, the line depth for Doppler broadening decreases *exponentially* as the wavelength moves away from the central wavelength λ_0 . This rapid decline is due to the high-speed exponential “tail” of the Maxwell–Boltzmann velocity distribution and is a much faster falloff in strength than for natural broadening.

Doppler shifts caused by the large-scale turbulent motion of large masses of gas (as opposed to the random motion of the individual atoms) can also be accommodated by Eq. (9.62) if the distribution of turbulent velocities follows the Maxwell–Boltzmann distribution. In that case,

$$(\Delta\lambda)_{1/2} = \frac{2\lambda}{c} \sqrt{\left(\frac{2kT}{m} + v_{\text{turb}}^2\right) \ln 2}, \quad (9.63)$$

where v_{turb} is the most probable turbulent speed. The effect of turbulence on line profiles is particularly important in the atmospheres of giant and supergiant stars. In

fact, the existence of turbulence in the atmospheres of these stars was first deduced from the inordinately large effect of Doppler broadening on their spectra.

Other sources of Doppler broadening involve orderly, coherent mass motions, such as stellar rotation, pulsation, and mass loss. These phenomena can have a substantial effect on the shape and width of the line profiles but cannot be combined with the results of Doppler broadening produced by random thermal motions obeying the Maxwell–Boltzmann distribution. For example, the characteristic P Cygni profile associated with mass loss will be discussed in Section 12.3 (see Fig. 12.17).

3. **Pressure (and collisional) broadening.** The orbitals of an atom can be perturbed in a collision with a neutral atom or by a close encounter involving the electric field of an ion. The results of individual collisions are called *collisional broadening*, and the statistical effects of the electric fields of large numbers of closely passing ions is termed *pressure broadening*; however, in the following discussion, both of these effects will be collectively referred to as pressure broadening. In either case, the outcome depends on the average time between collisions or encounters with other atoms and ions.

Calculating the precise width and shape of a pressure-broadened line is quite complicated. Atoms and ions of the same or different elements, as well as free electrons, are involved in these collisions and close encounters. The general shape of the line, however, is like that found for natural broadening, Eq. (9.61), and the line profile shared by natural and pressure broadening is sometimes referred to as a *damping profile* (also known as a *Lorentz profile*), so named because the shape is characteristic of the spectrum of radiation emitted by an electric charge undergoing damped simple harmonic motion. The values of the full width at half-maximum for natural and pressure broadening usually prove to be comparable, although the pressure profile can at times be more than an order of magnitude wider.

An estimate of pressure broadening due to collisions with atoms of a single element can be obtained by taking the value of Δt_0 in Eq. (9.61) to be the average time between collisions. This time is approximately equal to the mean free path between collisions divided by the average speed of the atoms. Using Eq. (9.12) for the mean free path and Eq. (8.2) for the speed, we find that

$$\Delta t_0 \approx \frac{\ell}{v} = \frac{1}{n\sigma\sqrt{2kT/m}},$$

where m is the mass of an atom, σ is its collision cross section, and n is the number density of the atoms. Thus the width of the spectral line due to pressure broadening is on the order of

$$\Delta\lambda = \frac{\lambda^2}{c} \frac{1}{\pi \Delta t_0} \approx \frac{\lambda^2}{c} \frac{n\sigma}{\pi} \sqrt{\frac{2kT}{m}}. \quad (9.64)$$

Note that the width of the line is proportional to the number density n of the atoms.

The physical reason for the Morgan–Keenan luminosity classes is now clear. The narrower lines observed for the more luminous giant and supergiant stars are due to

the lower number densities in their extended atmospheres. Pressure broadening (with the width of the line profile proportional to n) broadens the lines formed in the denser atmospheres of main-sequence stars, where collisions occur more frequently.

Example 9.5.3. Again, consider the hydrogen atoms in the Sun's photosphere, where the temperature is 5777 K and the number density of hydrogen atoms is about $1.5 \times 10^{23} \text{ m}^{-3}$. Then the pressure broadening of the H α line should be roughly

$$\Delta\lambda \approx 2.36 \times 10^{-5} \text{ nm},$$

which is comparable to the result for natural broadening found earlier. However, if the number density of the atoms in the atmosphere of a star is larger, the line width will be larger as well—more than an order of magnitude larger in some cases.

The Voigt Profile

The total line profile, called a **Voigt profile**, is due to the contributions of both the Doppler and damping profiles. The wider line profile for Doppler broadening dominates near the central wavelength λ_0 . Farther from λ_0 , however, the exponential decrease in the line depth for Doppler broadening means that there is a transition to a damping profile in the wings at a distance of about 1.8 times the Doppler value of $(\Delta\lambda)_{1/2}$ from the center of the line. Thus line profiles tend to have *Doppler cores* and *damping wings*. Figure 9.19 schematically shows the Doppler and damping line profiles.

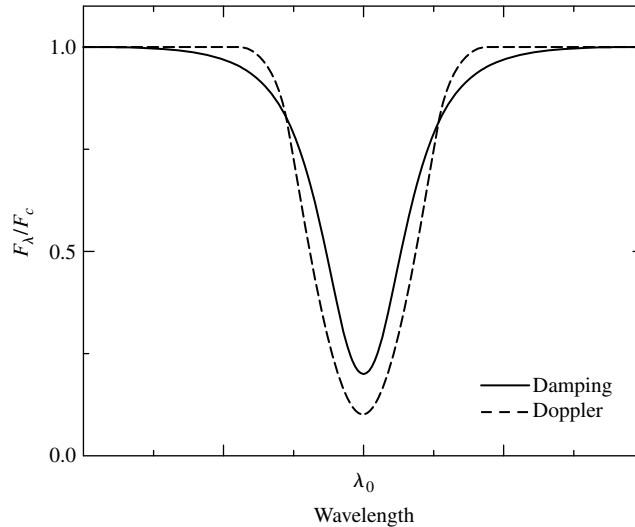


FIGURE 9.19 Schematic damping and Doppler line profiles, scaled so they have the same equivalent width.

Example 9.5.4. As a review of the ideas of spectral line formation discussed here and in Chapter 8, consider the subdwarfs of luminosity class VI or “sd,” which reside to the left of the main sequence (see Fig. 8.16). The spectra of these subdwarfs show that they are deficient in the atoms of metals (elements heavier than helium). Because ionized metals are an important source of electrons in stellar atmospheres, the electron number density is reduced. As mentioned in Section 8.1, fewer electrons with which ions may recombine means that a higher degree of ionization for all atoms can be achieved at the same temperature. Specifically, this reduces the number of H^- ions in the atmosphere by ionizing them, thereby diluting this dominant source of continuum opacity. As a consequence of a lower opacity, we can see longer distances into these stars before reaching an optical depth of $\tau_\lambda = 2/3$. The forest of metallic lines (which are already weakened by the low metal abundance of the subdwarfs) appears even weaker against the brighter continuum. Thus, as a result of an under-abundance of metals, the spectrum of a subdwarf appears to be that of a hotter and brighter star of earlier spectral type with less prominent metal lines (see Table 8.1). This is why it is more accurate to say that these stars are displaced to the *left* of the main sequence, toward higher temperatures, rather than one magnitude below the main sequence.

The simplest model used for calculating a line profile assumes that the star’s photosphere acts as a source of blackbody radiation and that the atoms above the photosphere remove photons from this continuous spectrum to form absorption lines. Although this **Schuster–Schwarzschild model** is inconsistent with the idea that photons of wavelength λ originate at an optical depth of $\tau_\lambda = 2/3$, it is still a useful approximation. In order to carry out the calculation, values for the temperature, density, and composition must be adopted for the region above the photosphere where the line is formed. The temperature and density determine the importance of Doppler and pressure broadening and are also used in the Boltzmann and Saha equations.

The calculation of a spectral line depends not only on the abundance of the element forming the line but also on the quantum-mechanical details of how atoms absorb photons. Let N be the number of atoms of a certain element lying *above a unit area* of the photosphere. N is a **column density** and has units of m^{-2} . (In other words, suppose a hollow tube with a cross section of 1 m^2 was stretched from the observer to the photosphere; the tube would then contain N atoms of the specified type.) To find the number of absorbing atoms per unit area, N_a , that have electrons in the proper orbital for absorbing a photon at the wavelength of the spectral line, the temperature and density are used in the Boltzmann and Saha equations to calculate the atomic states of excitation and ionization. Our goal is to determine the value of N_a by comparing the calculated and observed line profiles.

This task is complicated by the fact that not all transitions between atomic orbitals are equally likely. For example, an electron initially in the $n = 2$ orbital of hydrogen is about five times more likely to absorb an $\text{H}\alpha$ photon and make a transition to the $n = 3$ orbital than it is to absorb an $\text{H}\beta$ photon and jump to the $n = 4$ orbital. The relative probabilities of an electron making a transition from the same initial orbital are given by the *f-values* or *oscillator strengths* for that orbital. For hydrogen, $f = 0.637$ for the $\text{H}\alpha$ transition and $f = 0.119$ for $\text{H}\beta$. The oscillator strengths may be calculated numerically or measured in the laboratory, and they are defined so that the *f-values* for transitions from the same initial

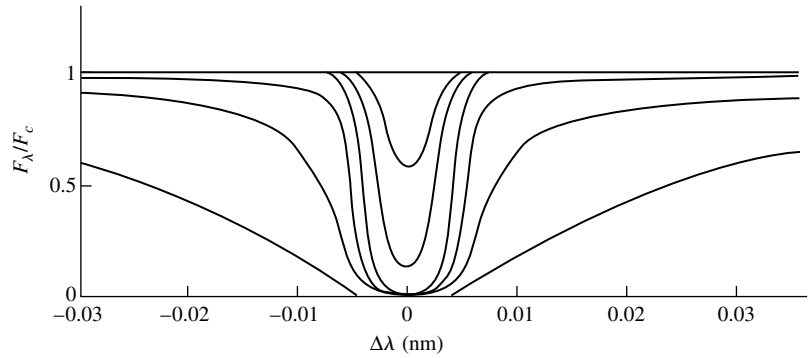


FIGURE 9.20 Voigt profiles of the K line of Ca II. The shallowest line is produced by $N_a = 3.4 \times 10^{15}$ ions m^{-2} , and the ions are ten times more abundant for each successively broader line. (Adapted from Novotny, *Introduction to Stellar Atmospheres and Interiors*, Oxford University Press, New York, 1973.)

orbital add up to the number of electrons in the atom or ion. Thus the oscillator strength is the effective number of electrons per atom participating in a transition, and so multiplying the number of absorbing atoms per unit area by the f -value gives the number of atoms lying above each square meter of the photosphere that are actively involved in producing a given spectral line, fN_a . Figure 9.20 shows the Voigt profiles of the K line of Ca II ($\lambda = 393.3$ nm) for various values of the number of absorbing calcium ions.

The Curve of Growth

The **curve of growth** is an important tool that astronomers use to determine the value of N_a and thus the abundances of elements in stellar atmospheres. As seen in Fig. 9.20, the equivalent width, W , of the line varies with N_a . A curve of growth, shown in Fig. 9.21, is a logarithmic graph of the equivalent width, W , as a function of the number of absorbing atoms, N_a . To begin with, imagine that a specific element is not present in a stellar atmosphere. As some of that element is introduced, a weak absorption line appears that is initially optically thin. If the number of the absorbing atoms is doubled, twice as much light is removed, and the equivalent width of the line is twice as great. So $W \propto N_a$, and the curve of growth is initially linear with $\ln N_a$. As the number of absorbing atoms continues to increase, the center of the line becomes optically thick as the maximum amount of flux at the line's center is absorbed.²⁹ With the addition of still more atoms, the line bottoms out and becomes saturated. The wings of the line, which are still optically thin, continue to deepen. This occurs with relatively little change in the line's equivalent width and produces a flattening on the curve of growth where $W \propto \sqrt{\ln N_a}$. Increasing the number of absorbing atoms still further increases the width of the pressure-broadening profile [recall Eq. (9.64)],

²⁹The zero flux at the center of the line shown in Fig. 9.20 is a peculiarity of the Schuster–Schwarzschild model. Actually, there is always *some* flux received at the central wavelength, λ_0 , even for very strong, optically thick lines. As a rule, the flux at any wavelength cannot fall below $F_\lambda = \pi S_\lambda$ ($\tau_\lambda = 2/3$), the value of the source function at an optical depth of $2/3$; see Problem 9.20.

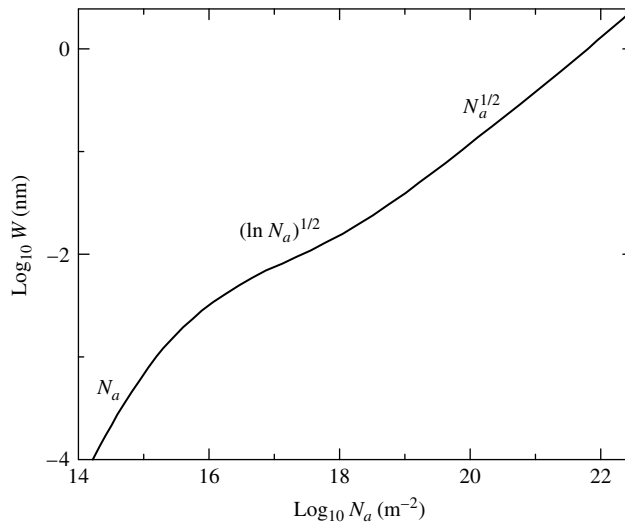


FIGURE 9.21 The curve of growth for the K line of Ca II. As N_a increases, the functional dependence of the equivalent width (W) changes. At various positions along the curve of growth, W is proportional to the functional forms indicated. (Figure adapted from Aller, *The Atmospheres of the Sun and Stars*, Ronald Press, New York, 1963.)

enabling it to contribute to the wings of the line. The equivalent width grows more rapidly, although not as steeply as at first, with approximately $W \propto \sqrt{N_a}$ for the total line profile. Using the curve of growth and a measured equivalent width, we can obtain the number of absorbing atoms. The Boltzmann and Saha equations are then used to convert this value into the total number of atoms of that element lying above the photosphere.

To reduce the errors involved in using a single spectral line, it is advantageous to locate, on a single curve of growth, the positions of the equivalent widths of several lines formed by transitions from the same initial orbital.³⁰ This can be accomplished by plotting $\log_{10}(W/\lambda)$ on the vertical axis and $\log_{10}[f N_a (\lambda/500 \text{ nm})]$ on the horizontal axis. This scaling results in a general curve of growth that can be used for several lines. Figure 9.22 shows a general curve of growth for the Sun. The use of such a curve of growth is best illustrated by an example.

Example 9.5.5. We will use Fig. 9.22 to find the number of sodium atoms above each square meter of the Sun's photosphere from measurements of the 330.238-nm and 588.997-nm absorption lines of sodium (Table 9.1). Values of $T = 5800 \text{ K}$ and $P_e = 1 \text{ N m}^{-2}$ were used for the temperature and electron pressure, respectively, to construct this curve of growth and will be adopted in the calculations that follow.

Both of these lines are produced when an electron makes an upward transition from the ground state orbital of the neutral Na I atom, and so these lines have the same value of N_a ,

³⁰This is just one of several possible ways of scaling the curve of growth. The assumptions used to obtain such a scaling are not valid for all broad lines (such as hydrogen) and may lead to inaccurate results.

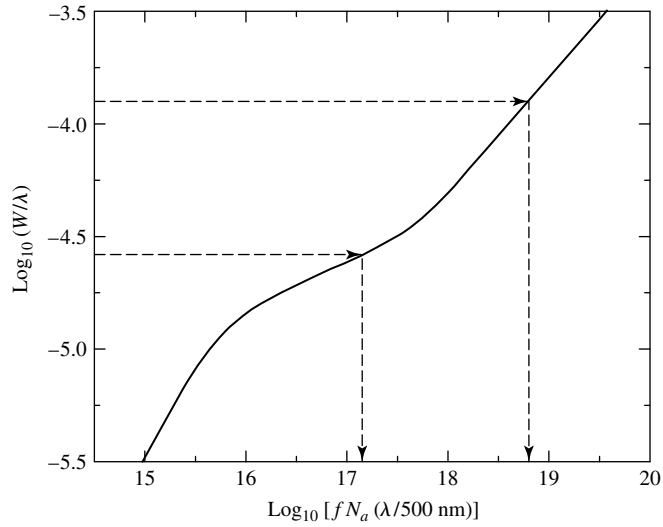


FIGURE 9.22 A general curve of growth for the Sun. The arrows refer to the data used in Example 9.5.5. (Figure adapted from Aller, *Atoms, Stars, and Nebulae*, Revised Edition, Harvard University Press, Cambridge, MA, 1971.)

TABLE 9.1 Data for Solar Sodium Lines. (From Aller, *Atoms, Stars, and Nebulae*, Revised Edition, Harvard University Press, Cambridge, MA, 1971.)

λ (nm)	W (nm)	f	$\log_{10}(W/\lambda)$	$\log_{10}[f(\lambda/500 \text{ nm})]$
330.238	0.0088	0.0214	-4.58	-1.85
588.997	0.0730	0.645	-3.90	-0.12

the number of absorbing sodium atoms per unit area above the continuum-forming layer of the photosphere. This number can be found using the values of $\log_{10}(W/\lambda)$ with the general curve of growth, Fig. 9.22, to obtain a value of $\log_{10}[fN_a(\lambda/500 \text{ nm})]$ for each line. The results are

$$\begin{aligned} \log_{10}\left(\frac{fN_a\lambda}{500 \text{ nm}}\right) &= 17.20 \quad \text{for the 330.238 nm line} \\ &= 18.83 \quad \text{for the 588.997 nm line.} \end{aligned}$$

To obtain the value of the number of absorbing atoms per unit area, N_a , we use the measured values of $\log_{10}[f(\lambda/500 \text{ nm})]$ together with

$$\log_{10} N_a = \log_{10}\left(\frac{fN_a\lambda}{500 \text{ nm}}\right) - \log_{10}\left(\frac{f\lambda}{500 \text{ nm}}\right),$$

to find

$$\log_{10} N_a = 17.15 - (-1.85) = 19.00 \quad \text{for the 330.238 nm line}$$

continued

and

$$\log_{10} N_a = 18.80 - (-0.12) = 18.92 \quad \text{for the 588.997 nm line.}$$

The average value of $\log_{10} N_a$ is 18.96; thus there are about 10^{19} Na I atoms in the ground state per square meter of the photosphere.

To find the total number of sodium atoms, the Boltzmann and Saha equations must be used; Eqs. (8.6) and (8.9), respectively. The difference in energy between the final and initial states [$E_b - E_a$ in Eq. (8.6)] is just the energy of the emitted photon. Using Eq. (5.3), the exponential term in the Boltzmann equation is

$$\begin{aligned} e^{-(E_b - E_a)/kT} &= e^{-hc/\lambda kT} \\ &= 5.45 \times 10^{-4} \quad \text{for the 330.238 nm line} \\ &= 1.48 \times 10^{-2} \quad \text{for the 588.997 nm line,} \end{aligned}$$

so nearly all of the neutral Na I atoms are in the ground state.

All that remains is to determine the total number of sodium atoms per unit area in all stages of ionization. If there are $N_I = 10^{19}$ neutral sodium atoms per square meter, then the number of singly ionized atoms, N_{II} , comes from the Saha equation:

$$\frac{N_{II}}{N_I} = \frac{2kT Z_{II}}{P_e Z_I} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_I/kT}.$$

Using $Z_I = 2.4$ and $Z_{II} = 1.0$ for the partition functions and $\chi_I = 5.14$ eV for the ionization energy of neutral sodium leads to $N_{II}/N_I = 2.43 \times 10^3$. There are about 2430 singly ionized sodium atoms for every neutral sodium atom in the Sun's photosphere,³¹ so the total number of sodium atoms per unit area above the photosphere is about

$$N = 2430 N_I = 2.43 \times 10^{22} \text{ m}^{-2}.$$

The mass of a sodium atom is 3.82×10^{-26} kg, so the mass of sodium atoms above each square meter of the photosphere is roughly 9.3×10^{-4} kg m⁻². (A more detailed analysis leads to a slightly lower value of 5.4×10^{-4} kg m⁻².) For comparison, the mass of hydrogen atoms per unit area is about 11 kg m⁻².

Thus the number of absorbing atoms can be determined by comparing the equivalent widths measured for different absorption lines produced by atoms or ions initially in the same state (and so having the same column density in the stellar atmosphere) with a theoretical curve of growth. A curve-of-growth analysis can also be applied to lines originating from atoms or ions in different initial states; then applying the Boltzmann equation to the relative numbers of atoms and ions in these different states of excitation allows the excitation temperature to be calculated. Similarly, it is possible to use the Saha equation to find either the electron pressure or the ionization temperature (if the other is known) in the atmosphere from the relative numbers of atoms at various stages of ionization.

³¹The ionization energy for Na II is 47.3 eV. This is sufficiently large to guarantee that $N_{III} \ll N_{II}$, so higher states of ionization can be neglected.

Computer Modeling of Stellar Atmospheres

The ultimate refinement in the analysis of stellar atmospheres is the construction of a *model atmosphere* on a computer. Each atmospheric layer is involved in the formation of line profiles and contributes to the spectrum observed for the star. All of the ingredients of the preceding discussion, plus the equations of hydrostatic equilibrium, thermodynamics, statistical and quantum mechanics, and the transport of energy by radiation and convection, are combined with extensive libraries of opacities to calculate how the temperature, pressure, and density vary with depth below the surface.³² These models not only provide details regarding line profiles; they also provide information about such fundamental properties as the effective temperature and surface gravity of the star. Only when the variables of the model have been “fine-tuned” to obtain good agreement with the observations can astronomers finally claim to have decoded the vast amount of information carried in the light from a star.

This basic procedure has led astronomers to an understanding of the abundances of the elements in the Sun (see Table 9.2) and other stars. Hydrogen and helium are by far the most common elements, followed by oxygen, carbon, and nitrogen; for every 10^{12} atoms of hydrogen, there are 10^{11} atoms of helium and about 10^9 atoms of oxygen. These figures are in very good agreement with abundances obtained from meteorites, giving astronomers

TABLE 9.2 The Most Abundant Elements in the Solar Photosphere. The relative abundance of an element is given by $\log_{10}(N_{\text{el}}/N_{\text{H}}) + 12$. (Data from Grevesse and Sauval, *Space Science Reviews*, 85, 161, 1998.)

Element	Atomic Number	Log Relative Abundance
Hydrogen	1	12.00
Helium	2	10.93 ± 0.004
Oxygen	8	8.83 ± 0.06
Carbon	6	8.52 ± 0.06
Neon	10	8.08 ± 0.06
Nitrogen	7	7.92 ± 0.06
Magnesium	12	7.58 ± 0.05
Silicon	14	7.55 ± 0.05
Iron	26	7.50 ± 0.05
Sulfur	16	7.33 ± 0.11
Aluminum	13	6.47 ± 0.07
Argon	18	6.40 ± 0.06
Calcium	20	6.36 ± 0.02
Sodium	11	6.33 ± 0.03
Nickel	28	6.25 ± 0.04

³²Details of the construction of a model star will be deferred to Chapter 10.

confidence in their results.³³ This knowledge of the basic ingredients of the universe provides invaluable observational tests and constraints for some of the most fundamental theories in astronomy: the nucleosynthesis of light elements as a result of stellar evolution, the production of heavier elements by supernovae, and the Big Bang that produced the primordial hydrogen and helium that started it all.

SUGGESTED READING

General

Hearnshaw, J. B., *The Analysis of Starlight*, Cambridge University Press, Cambridge, 1986.
Kaler, James B., *Stars and Their Spectra*, Cambridge University Press, Cambridge, 1997.

Technical

Aller, Lawrence H., *The Atmospheres of the Sun and Stars*, Ronald Press, New York, 1963.
Aller, Lawrence H., *Atoms, Stars, and Nebulae*, Third Edition, Cambridge University Press, New York, 1991.
Böhm-Vitense, Erika, “The Effective Temperature Scale,” *Annual Review of Astronomy and Astrophysics*, 19, 295, 1981.
Böhm-Vitense, Erika, *Stellar Astrophysics, Volume 2: Stellar Atmospheres*, Cambridge University Press, Cambridge, 1989.
Cox, Arthur N. (editor), *Allen’s Astrophysical Quantities*, Fourth Edition, AIP Press, New York, 2000.
Gray, David F., *The Observation and Analysis of Stellar Photospheres*, Third Edition, Cambridge University Press, Cambridge, 2005.
Grevesse, N., and Sauval, A. J., “Standard Solar Composition,” *Space Science Reviews*, 85, 161, 1998.
Iglesias, Carlos J., and Rogers, Forrest J., “Updated OPAL Opacities,” *The Astrophysical Journal*, 464, 943, 1996.
Mihalas, Dimitri, *Stellar Atmospheres*, Second Edition, W.H. Freeman, San Francisco, 1978.
Mihalas, Dimitri, and Weibel-Mihalas, Barbara, *Foundations of Radiation Hydrodynamics*, Dover Publications, Inc., Mineola, NY, 1999.
Novotny, Eva, *Introduction to Stellar Atmospheres and Interiors*, Oxford University Press, New York, 1973.
Rogers, Forrest, and Iglesias, Carlos, “The OPAL Opacity Code,”
<http://www-phys.llnl.gov/Research/OPAL/opal.html>.
Rybicki, George B., and Lightman, Alan P., *Radiative Processes in Astrophysics*, John Wiley and Sons, New York, 1979.

³³A notable exception is lithium, whose solar relative abundance of $10^{1.16}$ is significantly less than the value of $10^{3.31}$ obtained from meteorites. The efficient depletion of the Sun’s lithium, sparing only one of every 140 lithium atoms, is probably due to its destruction by nuclear reaction processes when the lithium is transported into the hot interior of the star by convection.

PROBLEMS

- 9.1** Evaluate the energy of the blackbody photons inside your eye. Compare this with the visible energy inside your eye while looking at a 100-W light bulb that is 1 m away. You can assume that the light bulb is 100% efficient, although in reality it converts only a few percent of its 100 watts into visible photons. Take your eye to be a hollow sphere of radius 1.5 cm at a temperature of 37°C. The area of the eye's pupil is about 0.1 cm². Why is it dark when you close your eyes?
- 9.2** (a) Find an expression for $n_\lambda d\lambda$, the number density of blackbody photons (the number of blackbody photons per m³) with a wavelength between λ and $\lambda + d\lambda$.
 (b) Find the total number of photons inside a kitchen oven set at 400°F (477 K), assuming a volume of 0.5 m³.
- 9.3** (a) Use the results of Problem 9.2 to find the total number density, n , of blackbody photons of all wavelengths. Also show that the average energy per photon, u/n , is

$$\frac{u}{n} = \frac{\pi^4 kT}{15(2.404)} = 2.70kT. \quad (9.65)$$

- (b) Find the average energy per blackbody photon at the center of the Sun, where $T = 1.57 \times 10^7$ K, and in the solar photosphere, where $T = 5777$ K. Express your answers in units of electron volts (eV).
- 9.4** Derive Eq. (9.11) for the blackbody radiation pressure.
- 9.5** Consider a spherical blackbody of radius R and temperature T . By integrating Eq. (9.8) for the radiative flux with $I_\lambda = B_\lambda$ over all outward directions, derive the Stefan–Boltzmann equation in the form of Eq. (3.17). (You will also have to integrate over all wavelengths and over the surface area of the sphere.)
- 9.6** Using the root-mean-square speed, v_{rms} , estimate the mean free path of the nitrogen molecules in your classroom at room temperature (300 K). What is the average time between collisions? Take the radius of a nitrogen molecule to be 0.1 nm and the density of air to be 1.2 kg m⁻³. A nitrogen molecule contains 28 nucleons (protons and neutrons).
- 9.7** Calculate how far you could see through Earth's atmosphere if it had the opacity of the solar photosphere. Use the value for the Sun's opacity from Example 9.2.2 and 1.2 kg m⁻³ for the density of Earth's atmosphere.
- 9.8** In Example 9.2.3, suppose that only two measurements of the specific intensity, I_1 and I_2 , are available, made at angles θ_1 and θ_2 . Determine expressions for the intensity $I_{\lambda,0}$ of the light above Earth's atmosphere and for the vertical optical depth of the atmosphere, $\tau_{\lambda,0}$, in terms of these two measurements.
- 9.9** Use the laws of conservation of relativistic energy and momentum to prove that an isolated electron cannot absorb a photon.
- 9.10** By measuring the slope of the curves in Fig. 9.10, verify that the decline of the curves after the peak in the opacity follows a Kramers law, $\bar{\kappa} \propto T^{-n}$, where $n \approx 3.5$.
- 9.11** According to one model of the Sun, the central density is 1.53×10^5 kg m⁻³ and the Rosseland mean opacity at the center is 0.217 m² kg⁻¹.
 (a) Calculate the mean free path of a photon at the center of the Sun.

- (b) Calculate the average time it would take for the photon to escape from the Sun if this mean free path remained constant for the photon's journey to the surface. (Ignore the fact that identifiable photons are constantly destroyed and created through absorption, scattering, and emission.)
- 9.12 If the temperature of a star's atmosphere is *increasing* outward, what type of spectral lines would you expect to find in the star's spectrum at those wavelengths where the opacity is greatest?
- 9.13 Consider a large hollow spherical shell of hot gas surrounding a star. Under what circumstances would you see the shell as a glowing *ring* around the star? What can you say about the optical thickness of the shell?
- 9.14 Verify that the emission coefficient, j_λ , has units of $\text{m s}^{-3} \text{sr}^{-1}$.
- 9.15 Derive Eq. (9.35) in Example 9.4.1, which shows how the intensity of a light ray is converted from its initial intensity I_λ to the value S_λ of the source function.
- 9.16 The transfer equation, Eq. (9.34), is written in terms of the distance, s , measured along the path of a light ray. In different coordinate systems, the transfer equation will look slightly different, and care must be taken to include all of the necessary terms.
- (a) Show that in a spherical coordinate system, with the center of the star at the origin, the transfer equation has the form

$$-\frac{\cos \theta'}{\kappa_\lambda \rho} \frac{dI_\lambda}{dr} = I_\lambda - S_\lambda,$$

where θ' is the angle between the ray and the outward radial direction. Note that you cannot simply replace s with r !

- (b) Use this form of the transfer equation to derive Eq. (9.31).
- 9.17 For a plane-parallel atmosphere, show that the Eddington approximation leads to expressions for the mean intensity, radiative flux, and radiation pressure given by Eqs. (9.46–9.48).
- 9.18 Using the Eddington approximation for a plane-parallel atmosphere, determine the values of I_{in} and I_{out} as functions of the vertical optical depth. At what depth is the radiation isotropic to within 1%?
- 9.19 Using the results for the plane-parallel gray atmosphere in LTE, determine the ratio of the effective temperature of a star to its temperature at the top of the atmosphere. If $T_e = 5777 \text{ K}$, what is the temperature at the top of the atmosphere?
- 9.20 Show that for a plane-parallel gray atmosphere in LTE, the (constant) value of the radiative flux is equal to π times the source function evaluated at an optical depth of $2/3$:

$$F_{\text{rad}} = \pi S(\tau_v = 2/3).$$

This function, called the **Eddington–Barbier relation**, says that the radiative flux received from the surface of the star is determined by the value of the source function at $\tau_v = 2/3$.

- 9.21 Consider a horizontal plane-parallel slab of gas of thickness L that is maintained at a constant temperature T . Assume that the gas has optical depth $\tau_{\lambda,0}$, with $\tau_\lambda = 0$ at the top surface of the slab. Assume further that no radiation enters the gas from outside. Use the general solution of the transfer equation (9.54) to show that when looking at the slab from above, you see blackbody radiation if $\tau_{\lambda,0} \gg 1$ and emission lines (where j_λ is large) if $\tau_{\lambda,0} \ll 1$. You may assume that the source function, S_λ , does not vary with position inside the gas. You may also assume thermodynamic equilibrium when $\tau_{\lambda,0} \gg 1$.